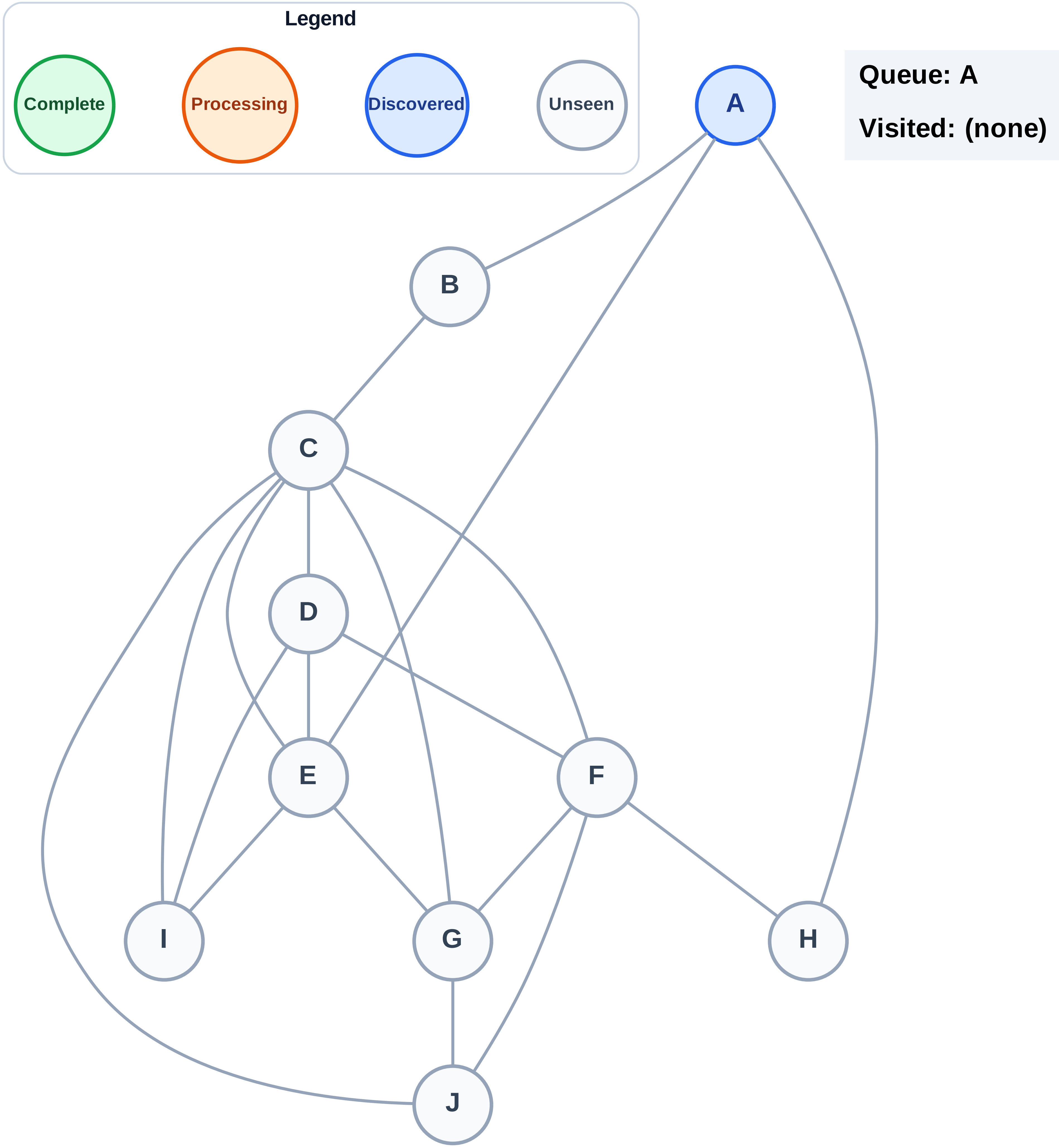
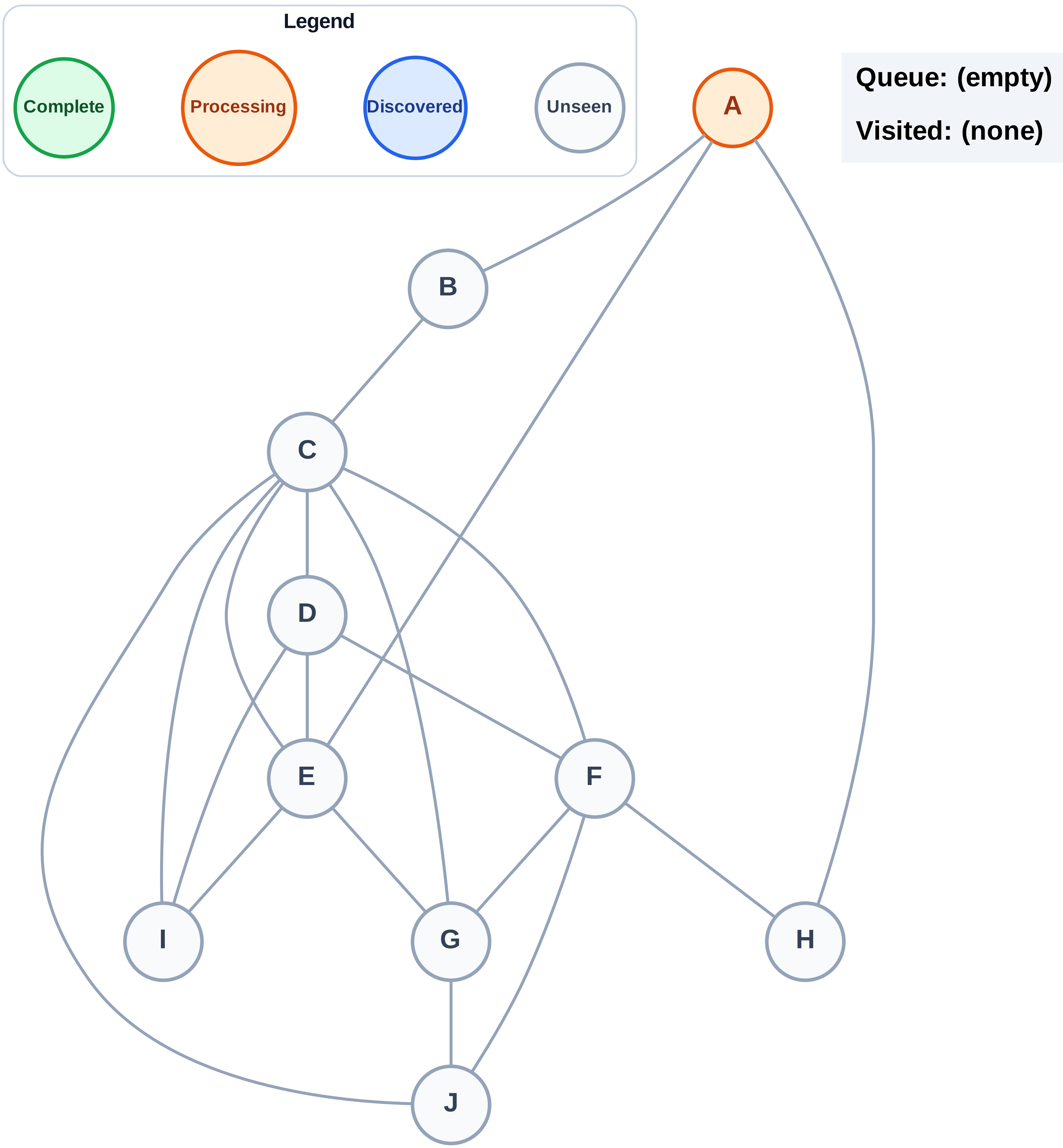
BFS Traversal

Step 1: Start at A



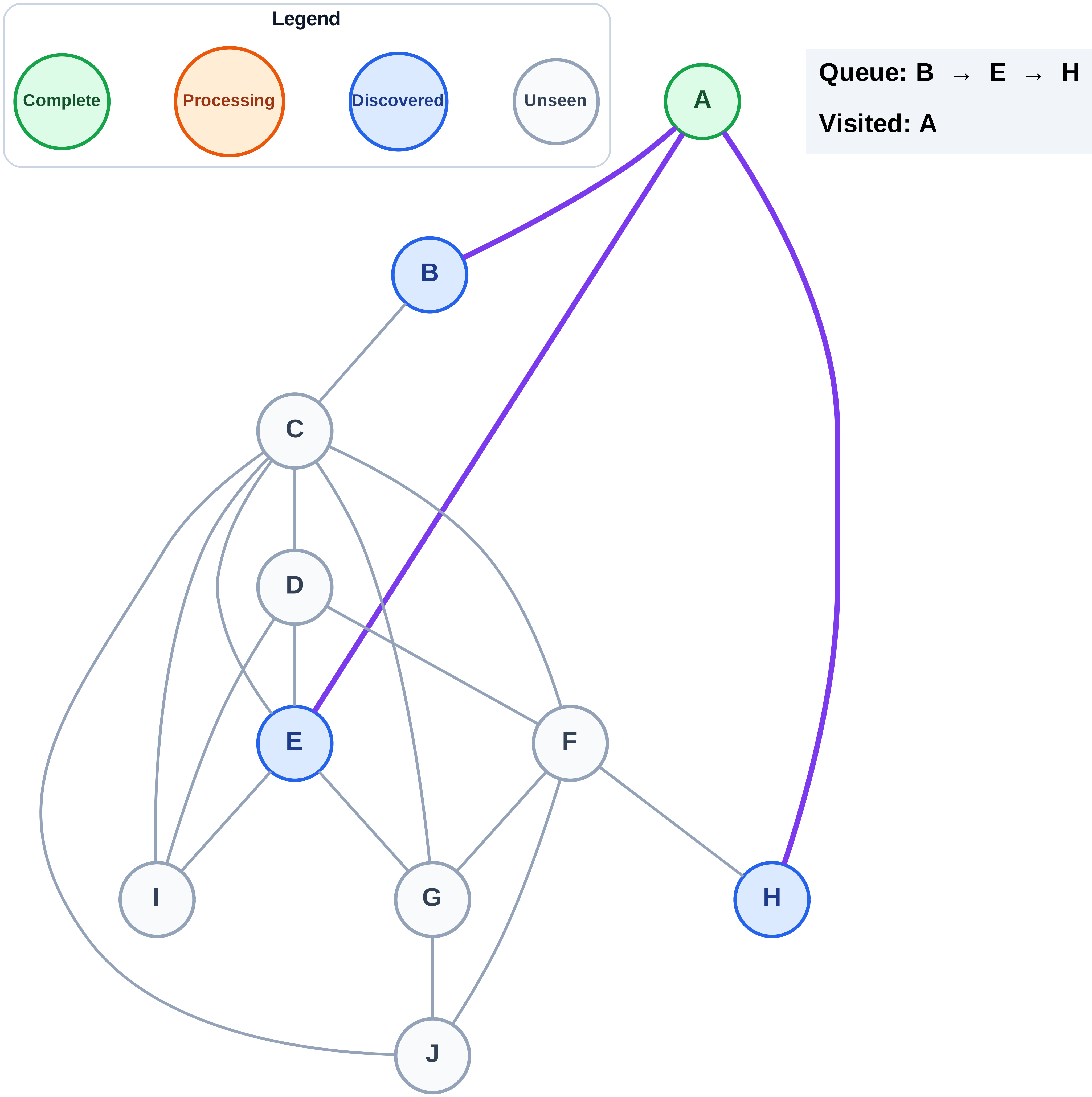
BFS Traversal

Step 2: Process node A



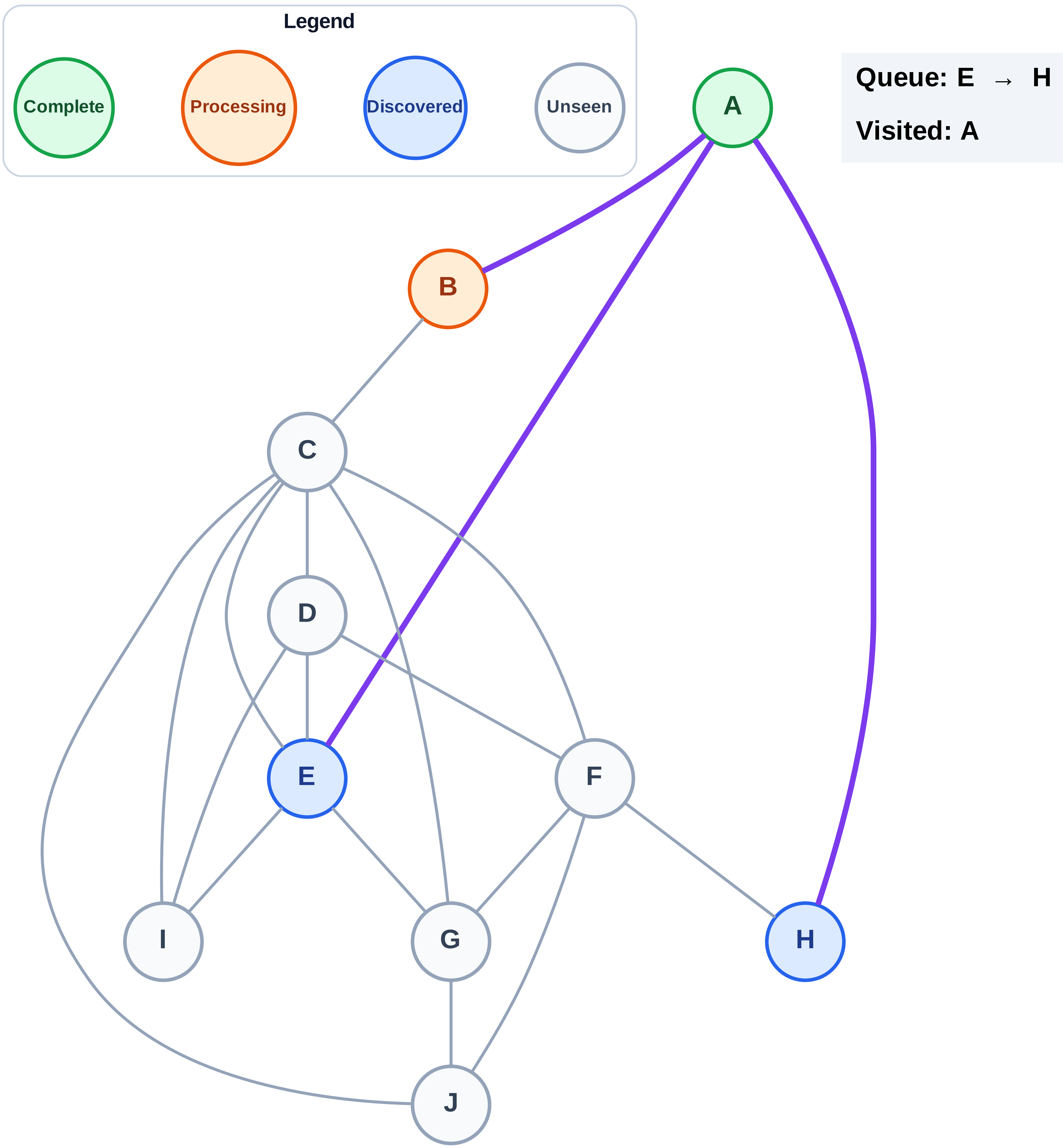
BFS Traversal

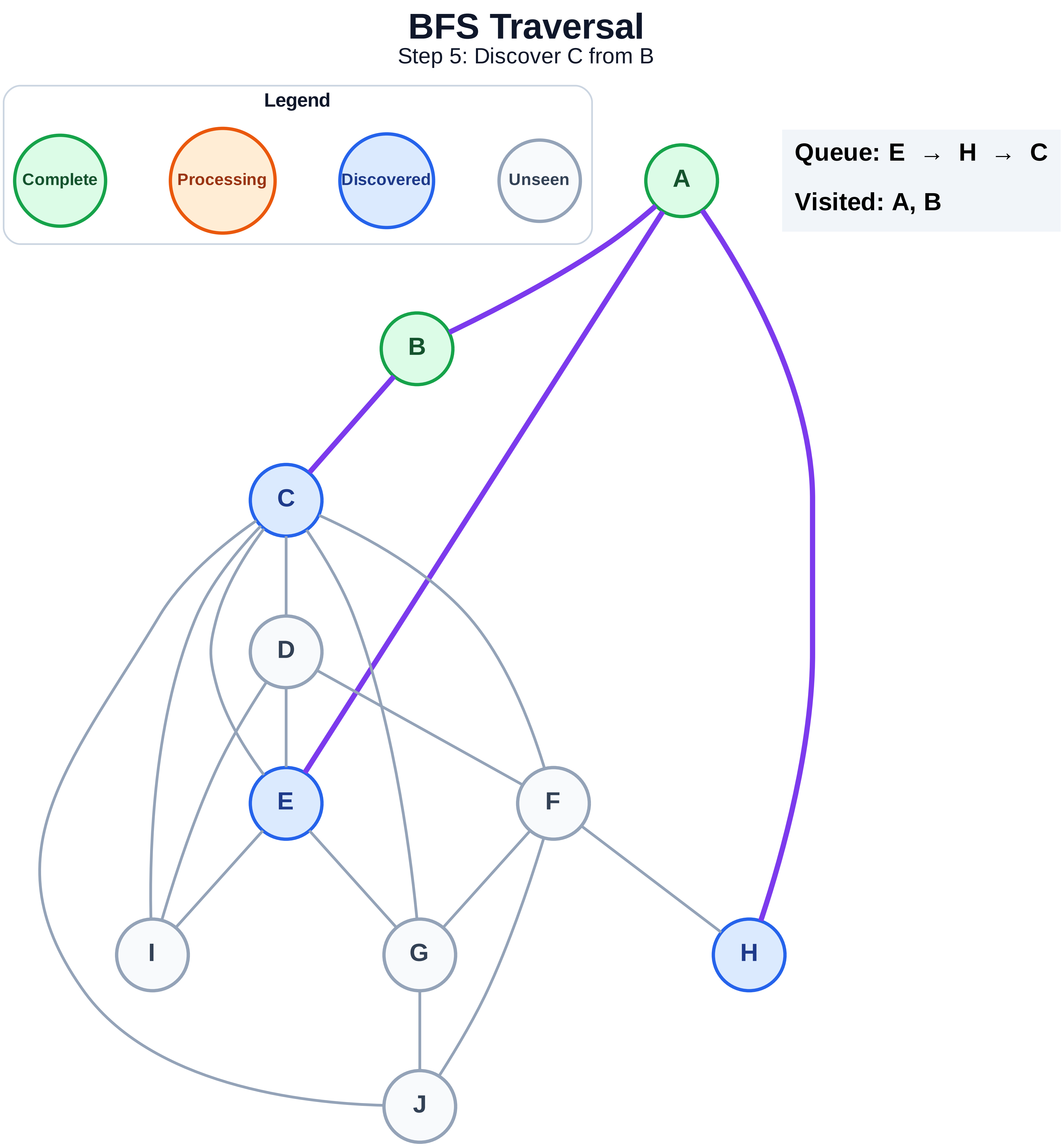
Step 3: Discover B, E, H from A



BFS Traversal

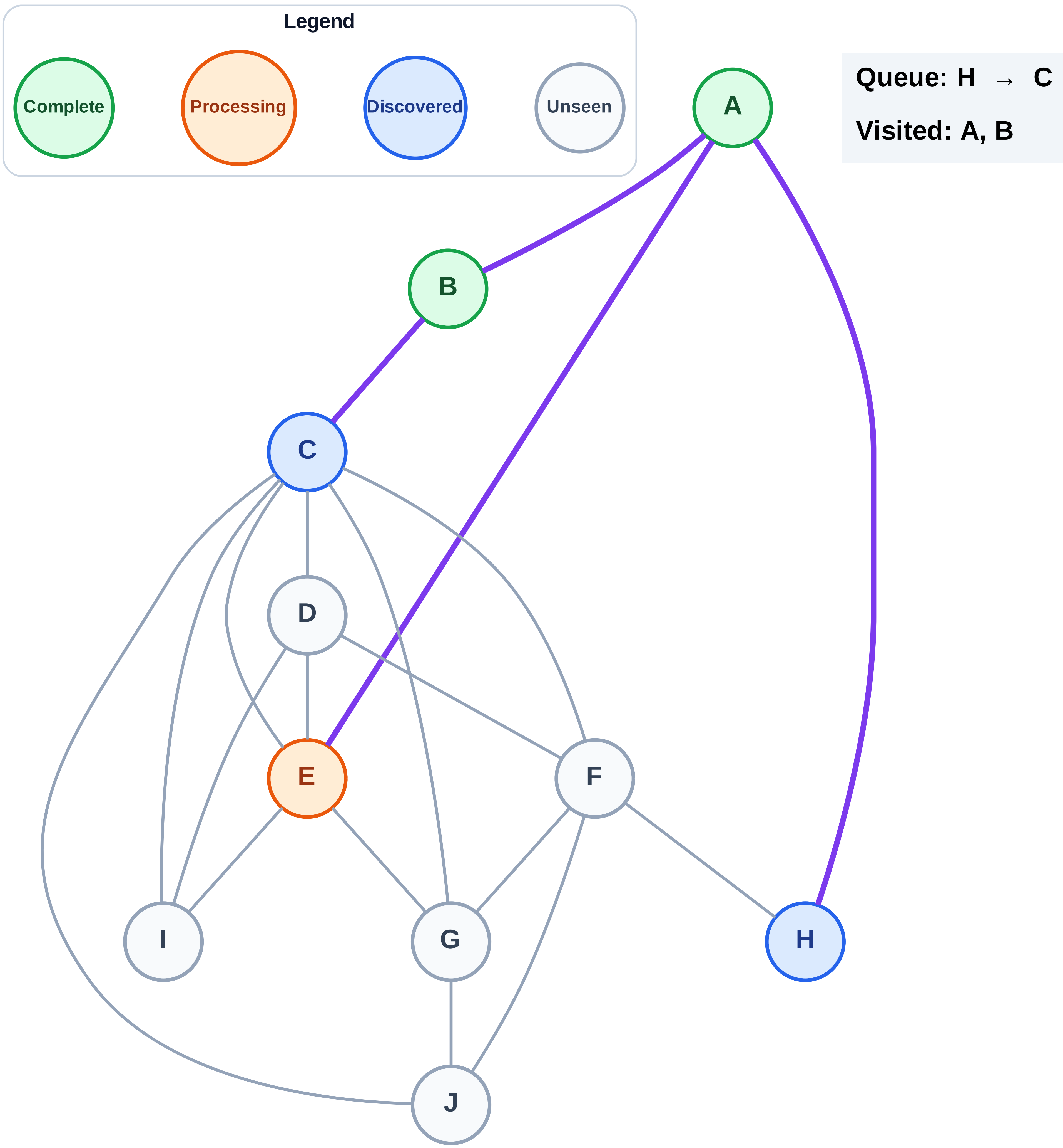
Step 4: Process node B




BFS Traversal
Step 5: Discover C from B
Legend
Complete
Processing
Discovered
Unseen
Queue: E → H → C
Visited: A, B
A
B
C
D
E
F
G
H
I
J

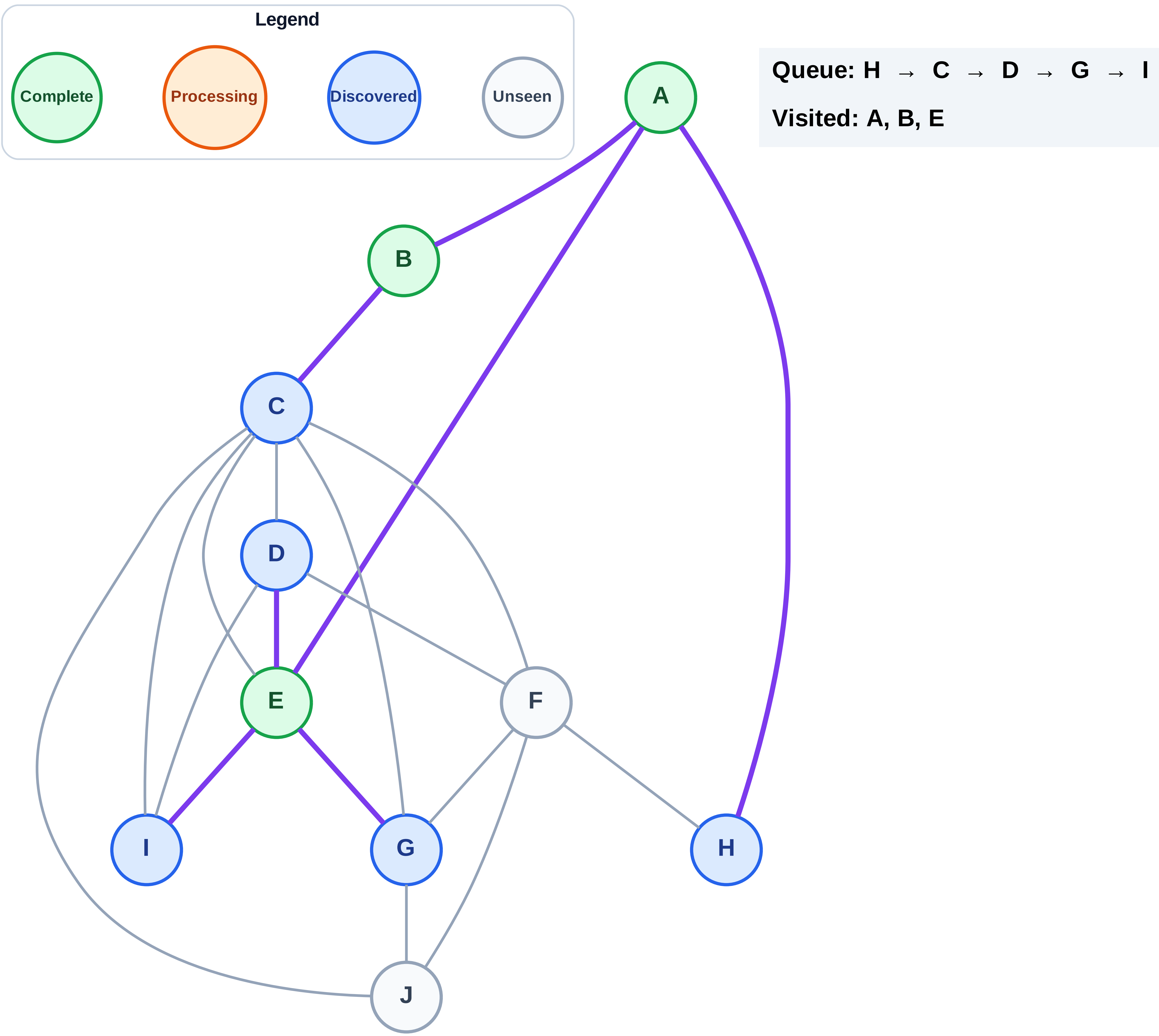
BFS Traversal

Step 6: Process node E



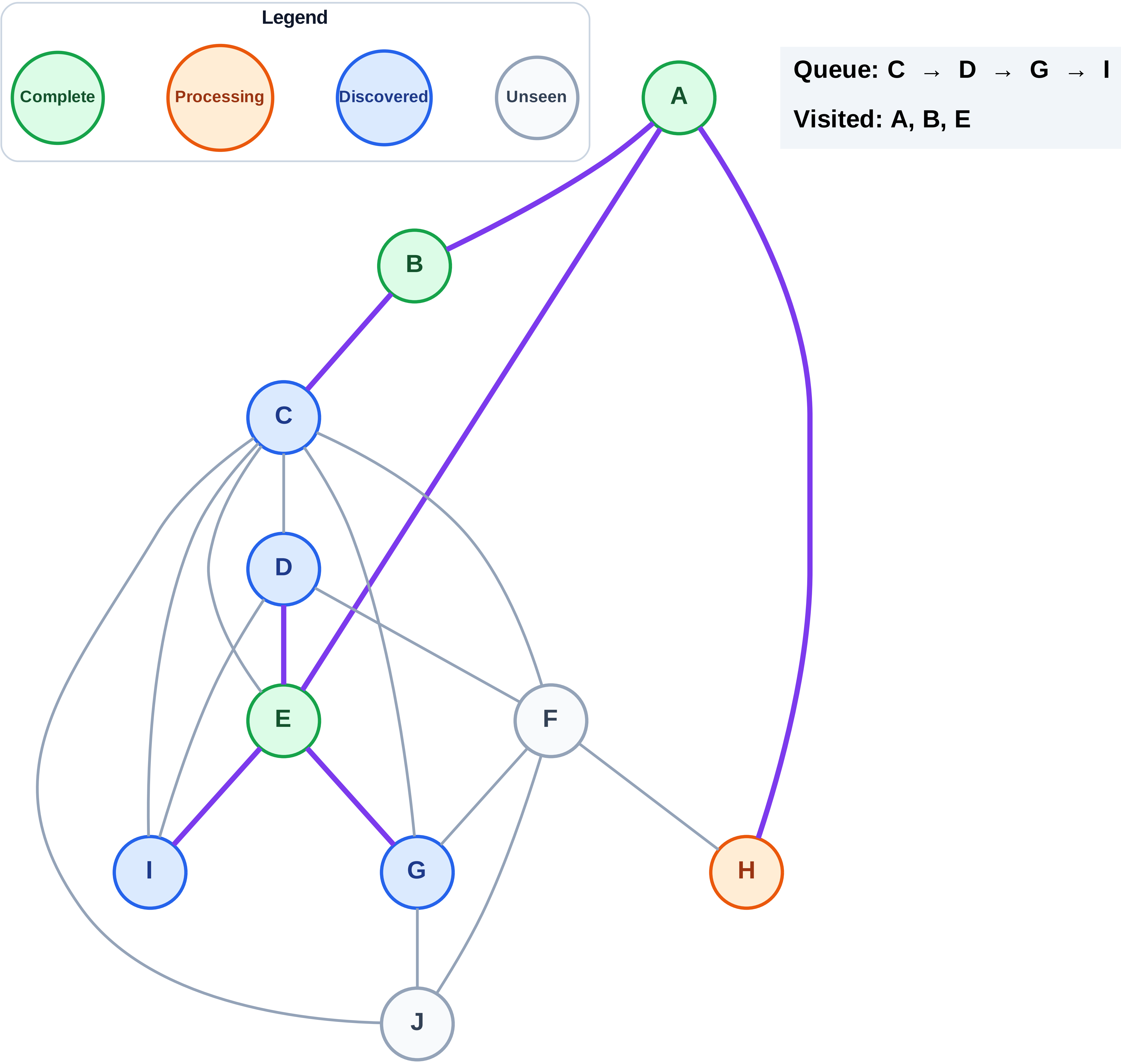
BFS Traversal

Step 7: Discover D, G, I from E



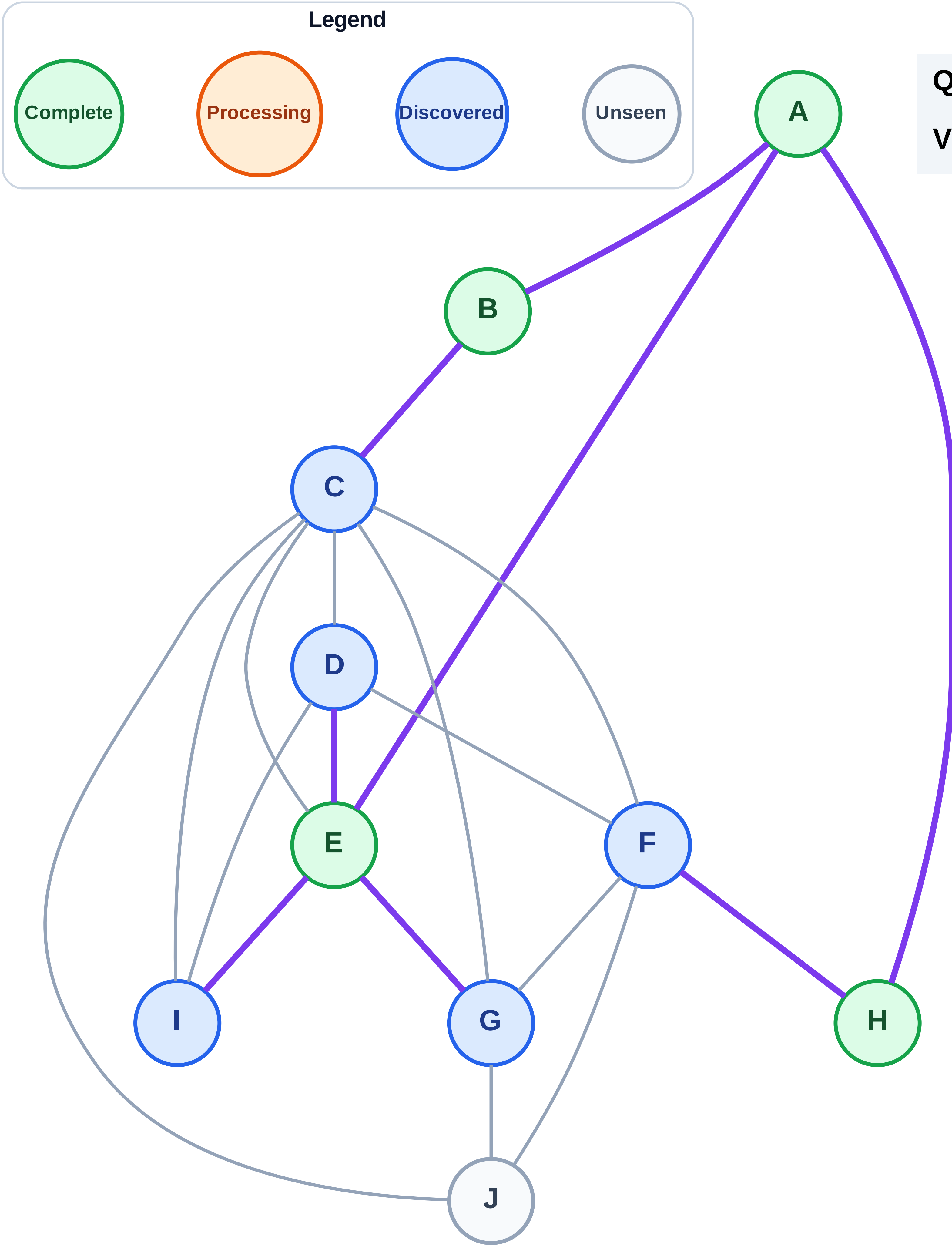
BFS Traversal

Step 8: Process node H



BFS Traversal

Step 9: Discover F from H

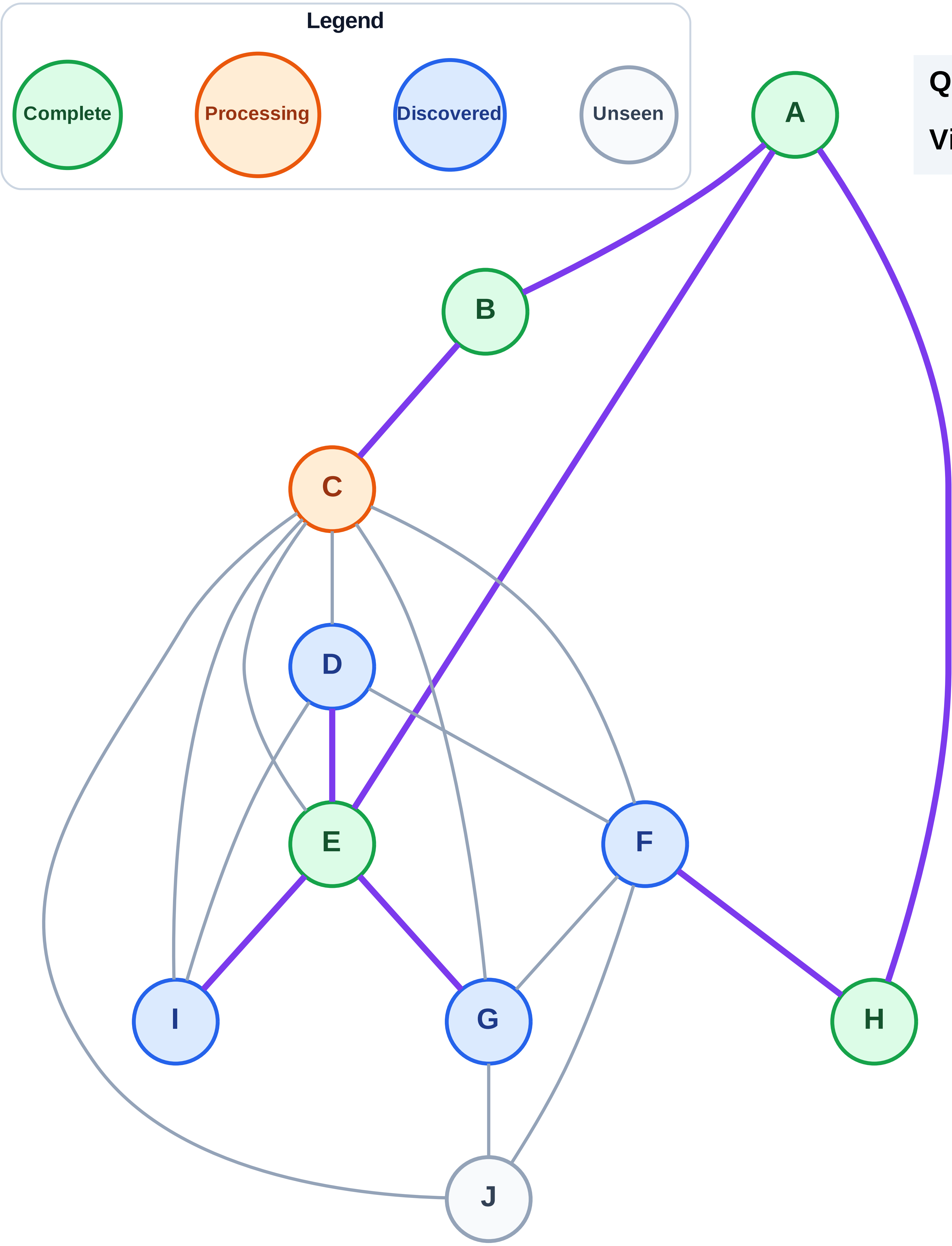


Queue: C → D → G → I → F

Visited: A, B, E, H

BFS Traversal

Step 10: Process node C

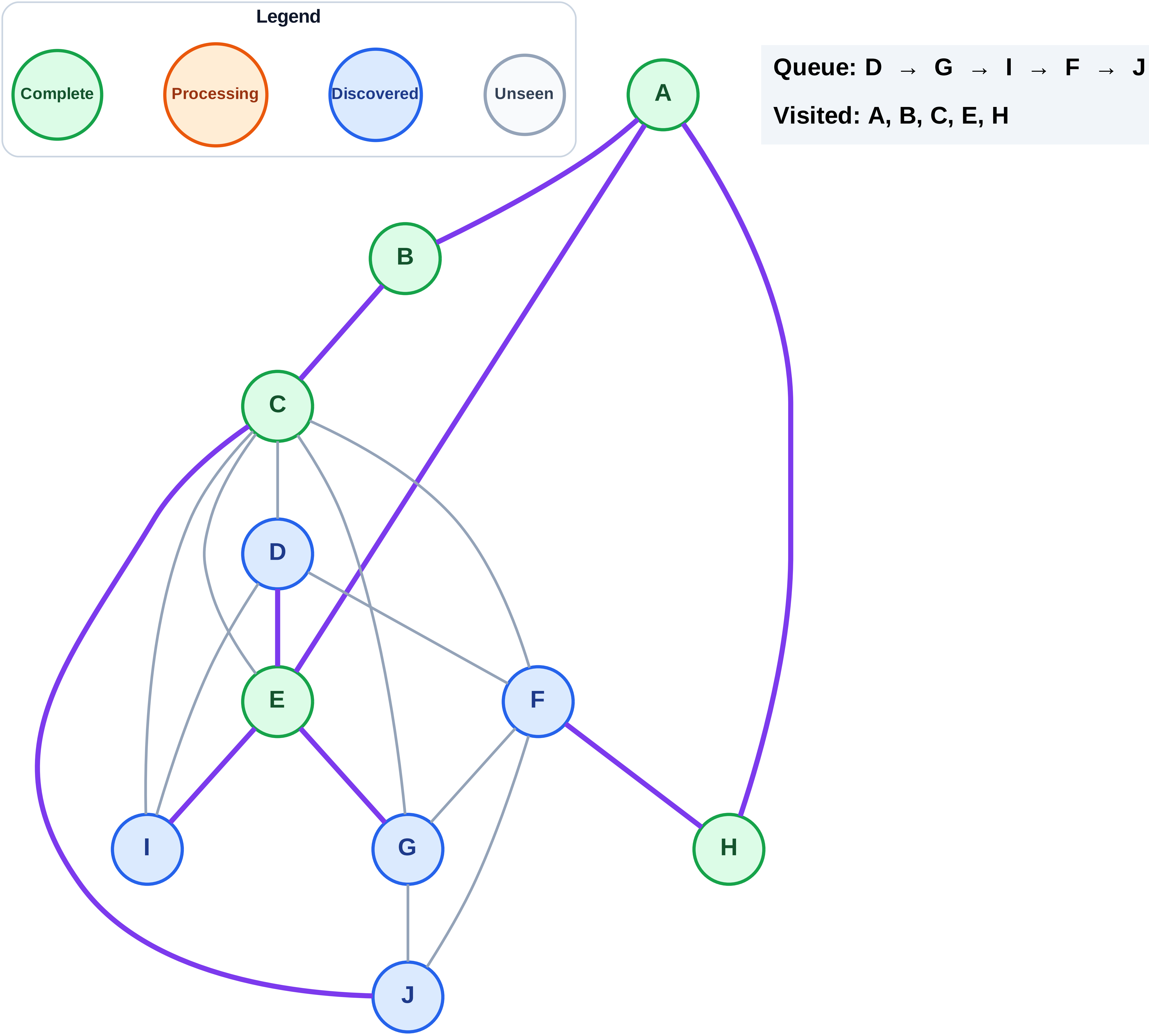


Queue: D → G → I → F

Visited: A, B, E, H

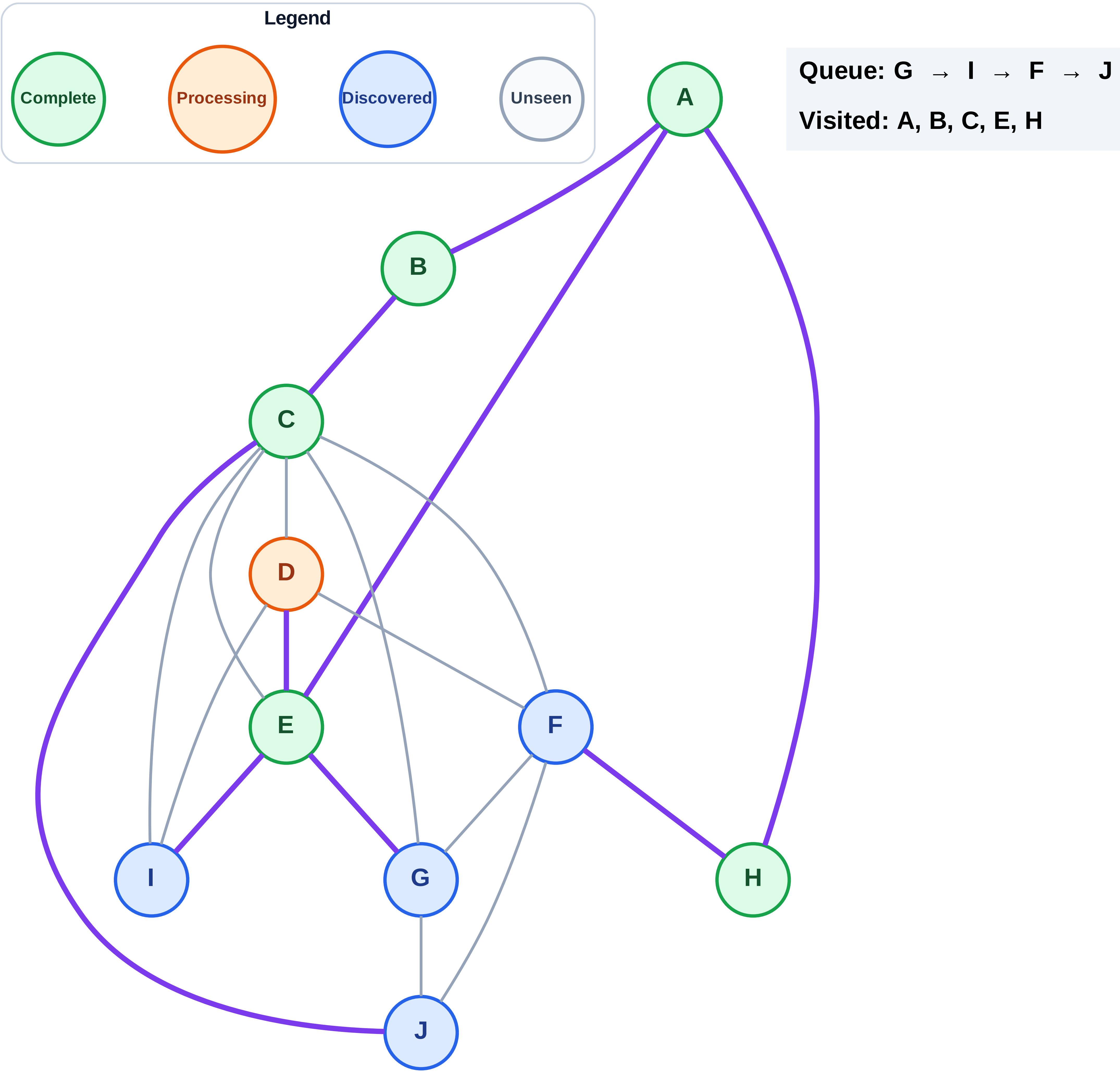
BFS Traversal

Step 11: Discover J from C



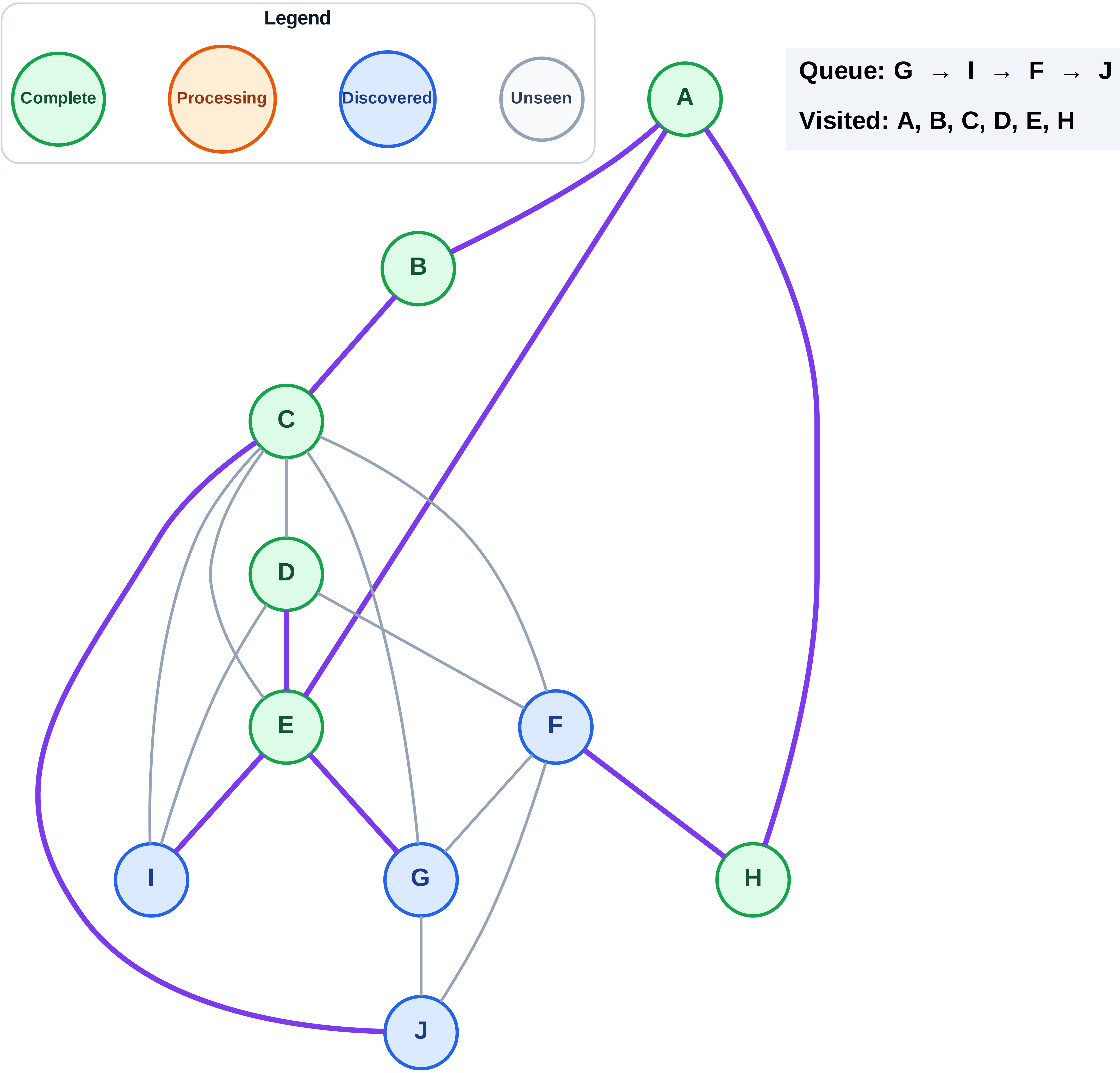
BFS Traversal

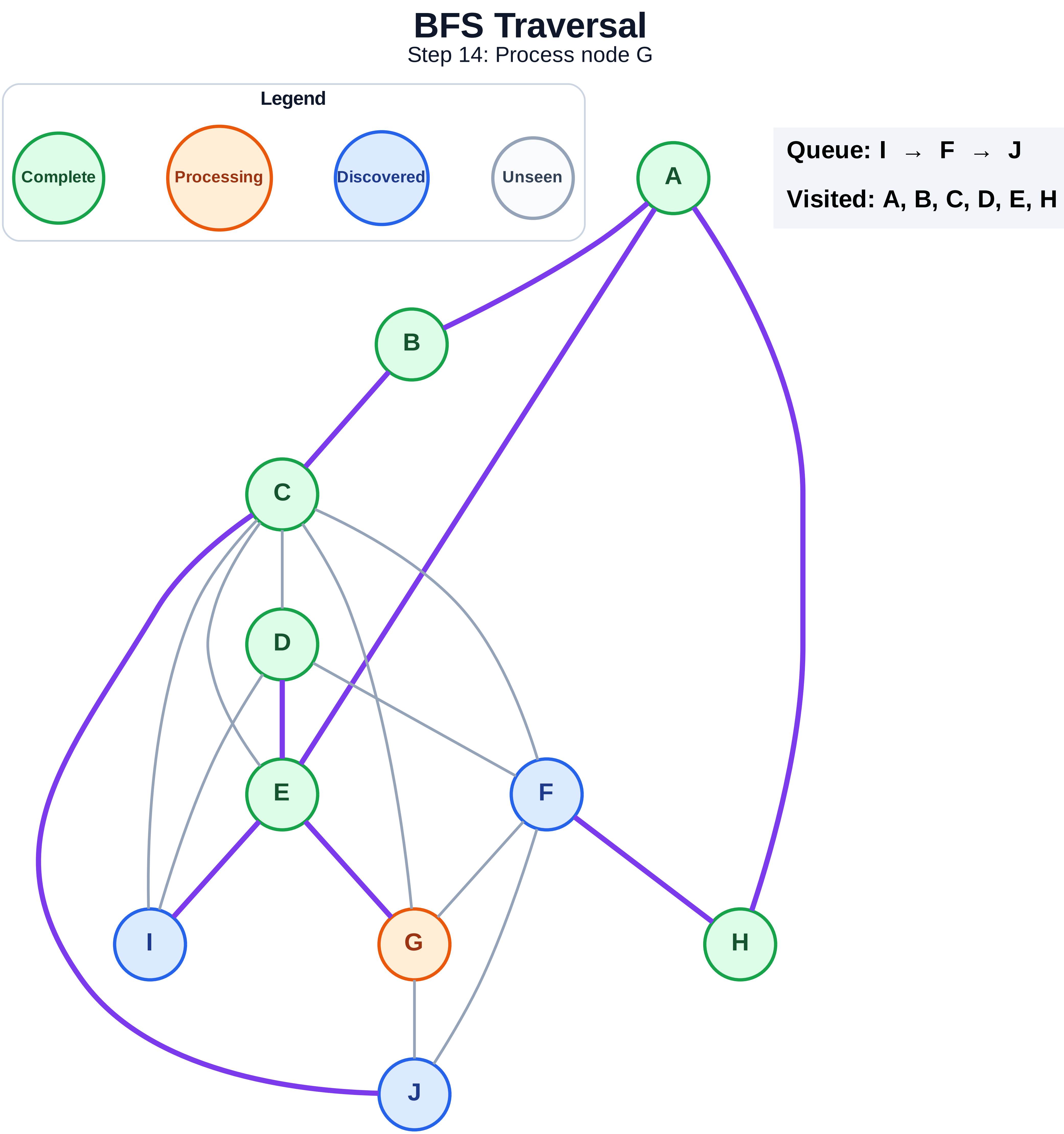
Step 12: Process node D



BFS Traversal

Step 13: No new neighbors from D

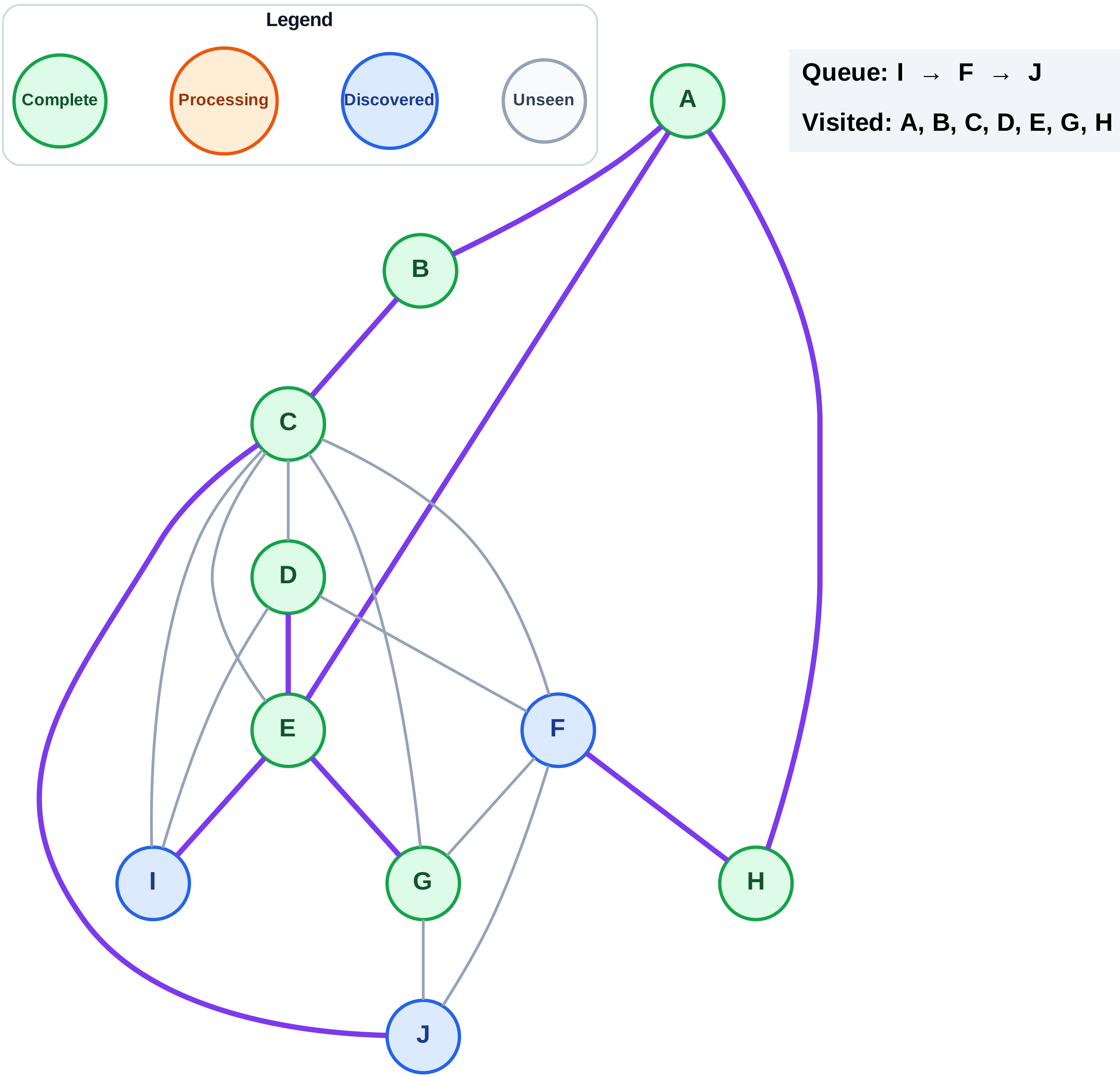




BFS Traversal
Step 14: Process node G
Legend
Complete
Processing
Discovered
Unseen
Queue: I → F → J
Visited: A, B, C, D, E, H
A
B
C
D
E
F
G
H
I
J

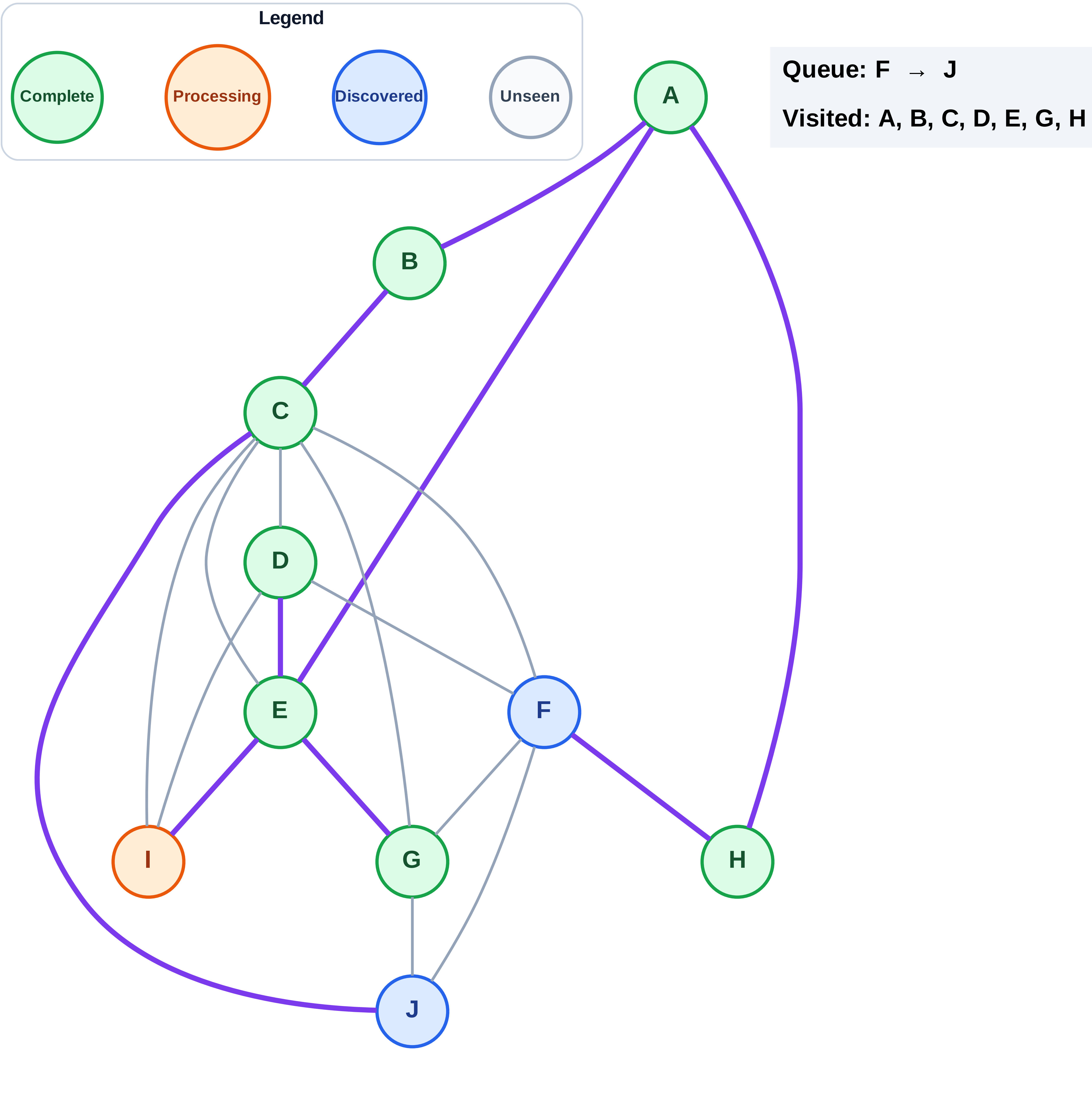
BFS Traversal

Step 15: No new neighbors from G



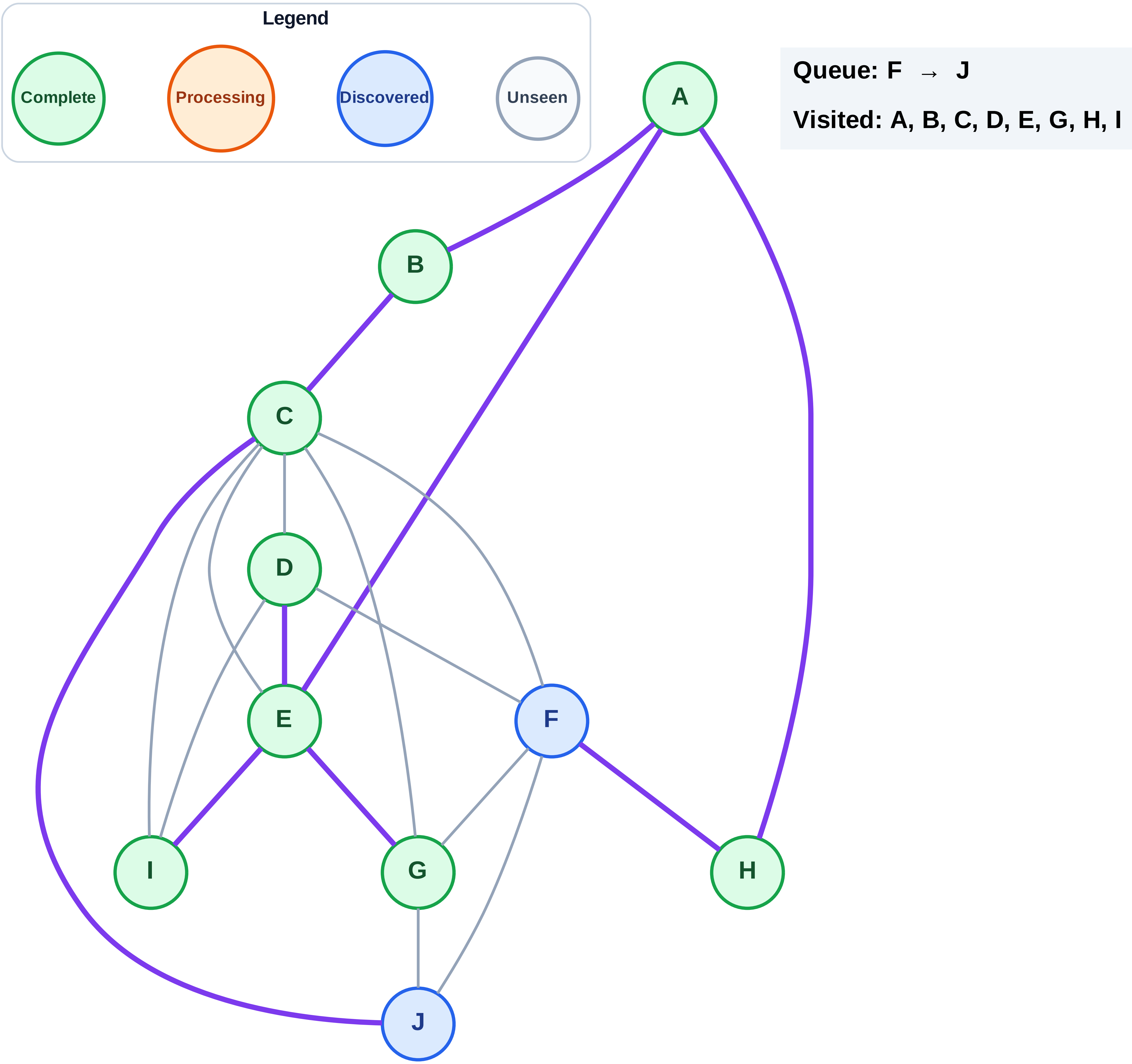
BFS Traversal

Step 16: Process node I



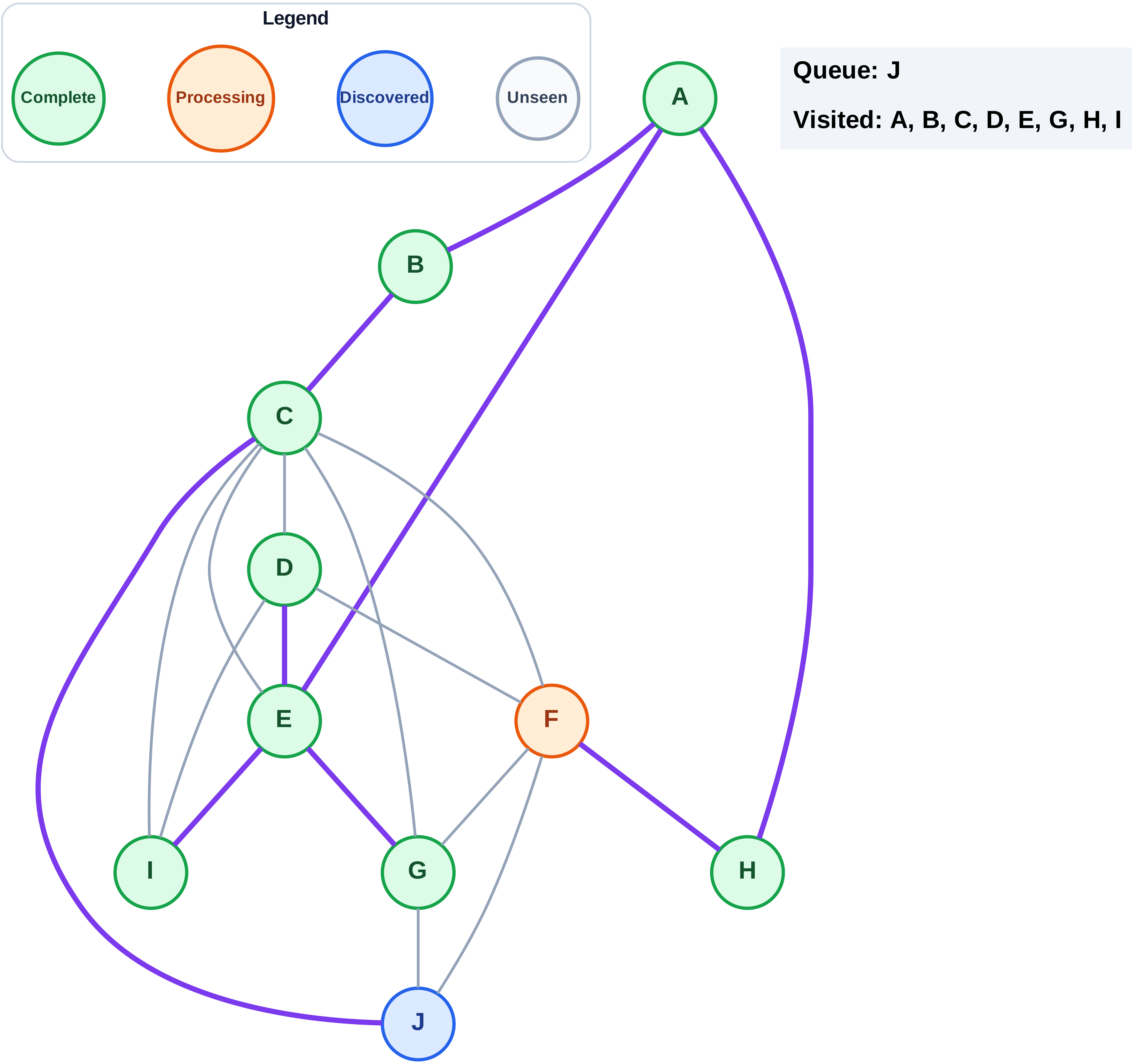
BFS Traversal

Step 17: No new neighbors from I



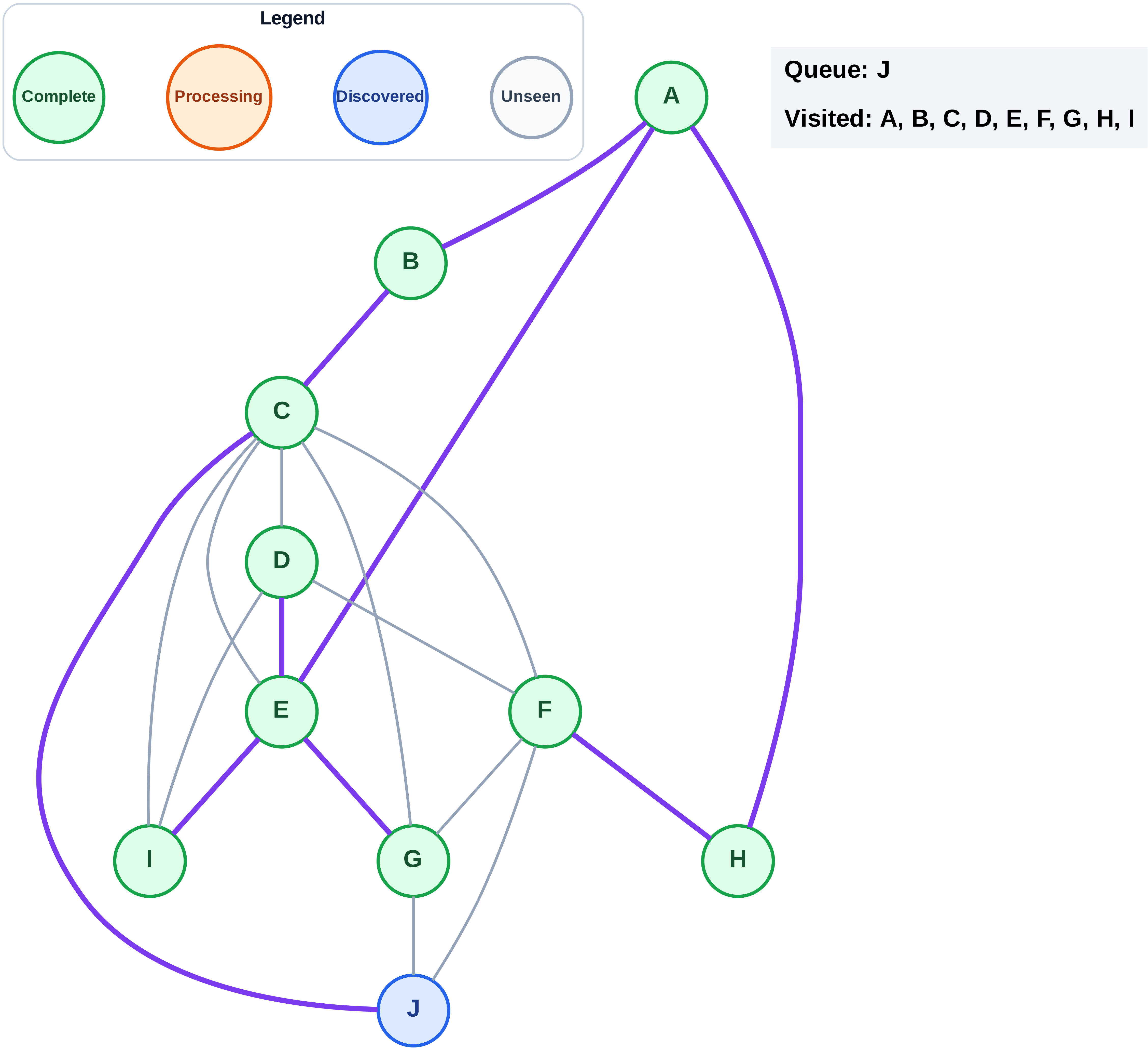
BFS Traversal

Step 18: Process node F



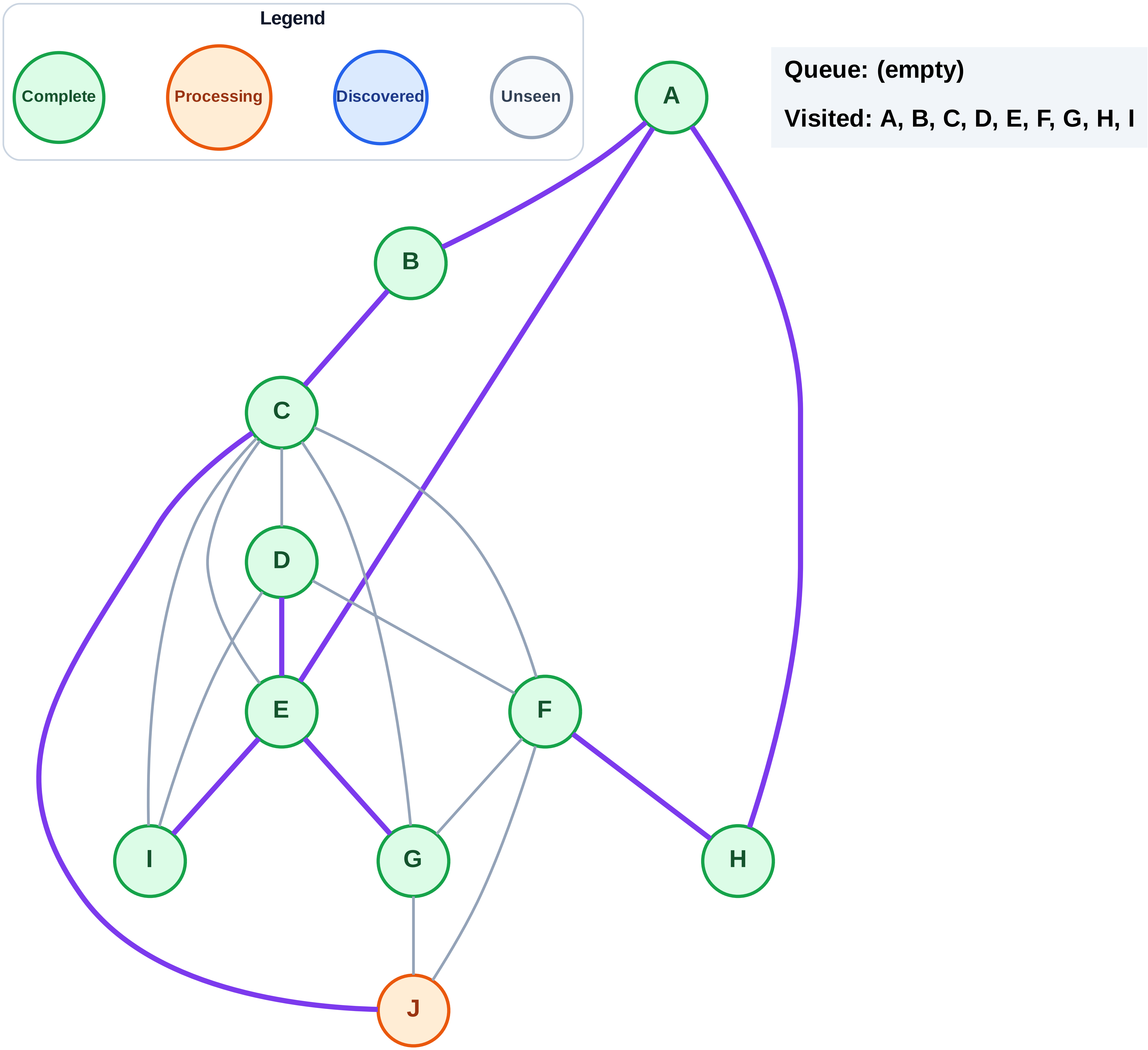
BFS Traversal

Step 19: No new neighbors from F



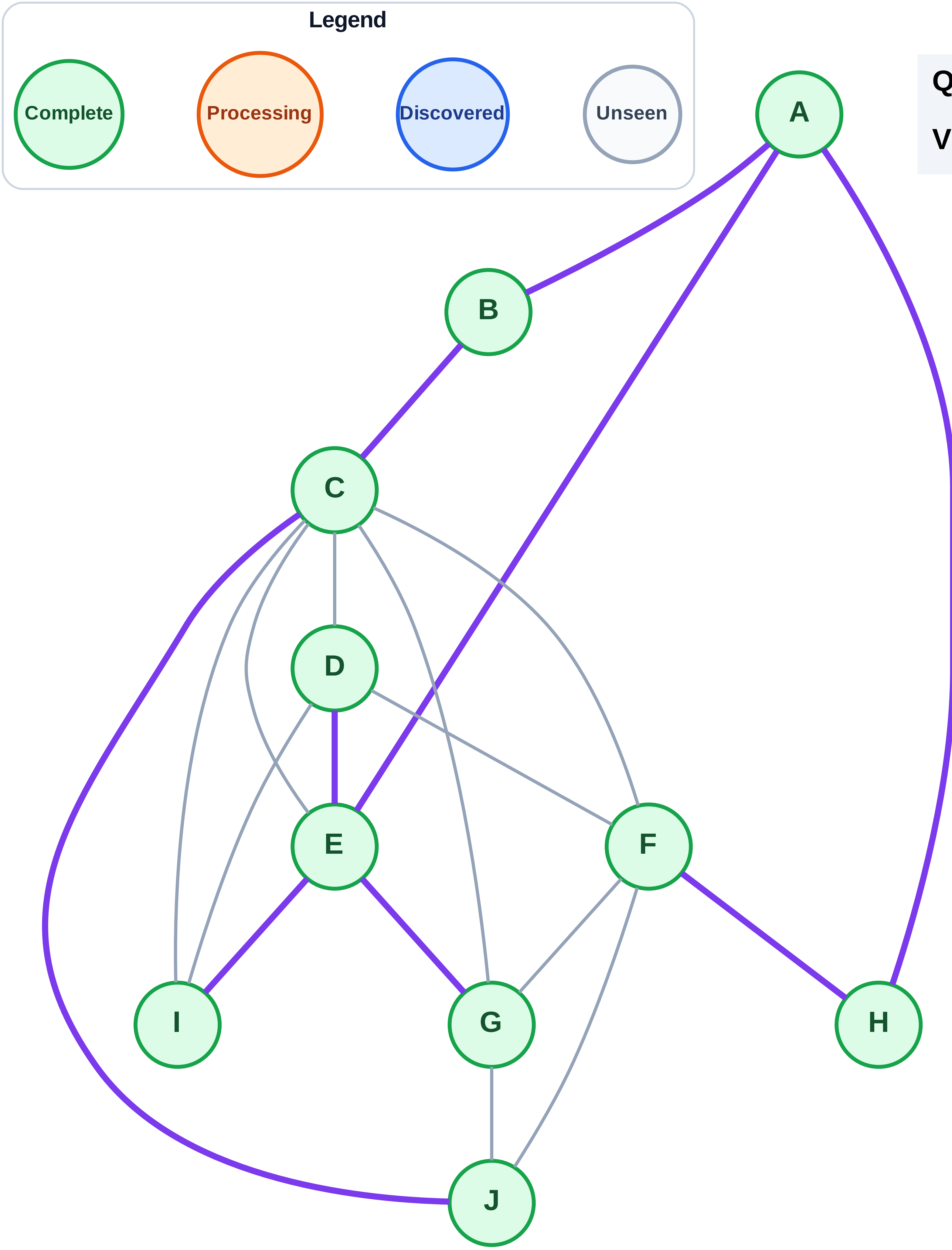
BFS Traversal

Step 20: Process node J



BFS Traversal

Step 21: No new neighbors from J

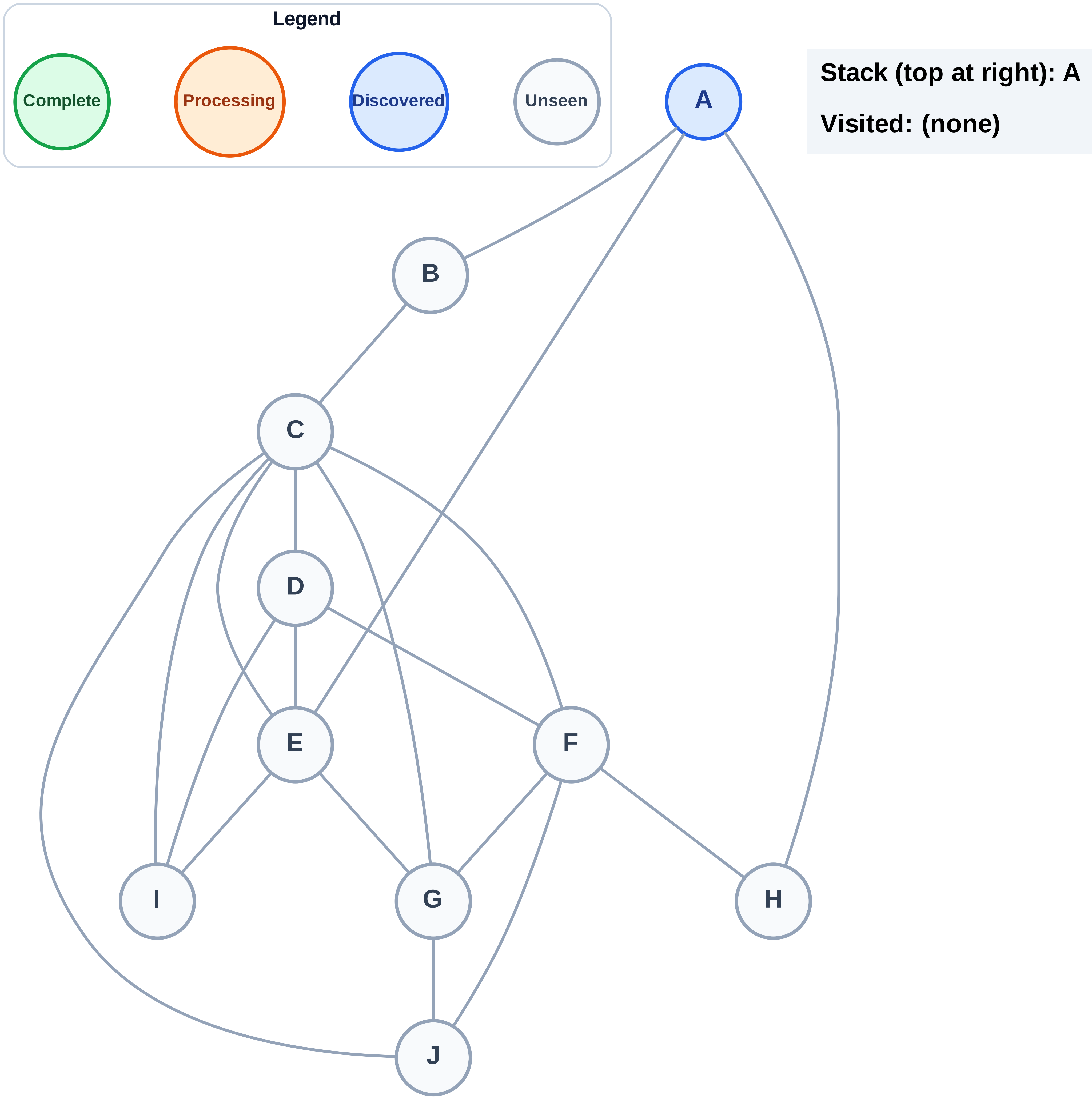


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

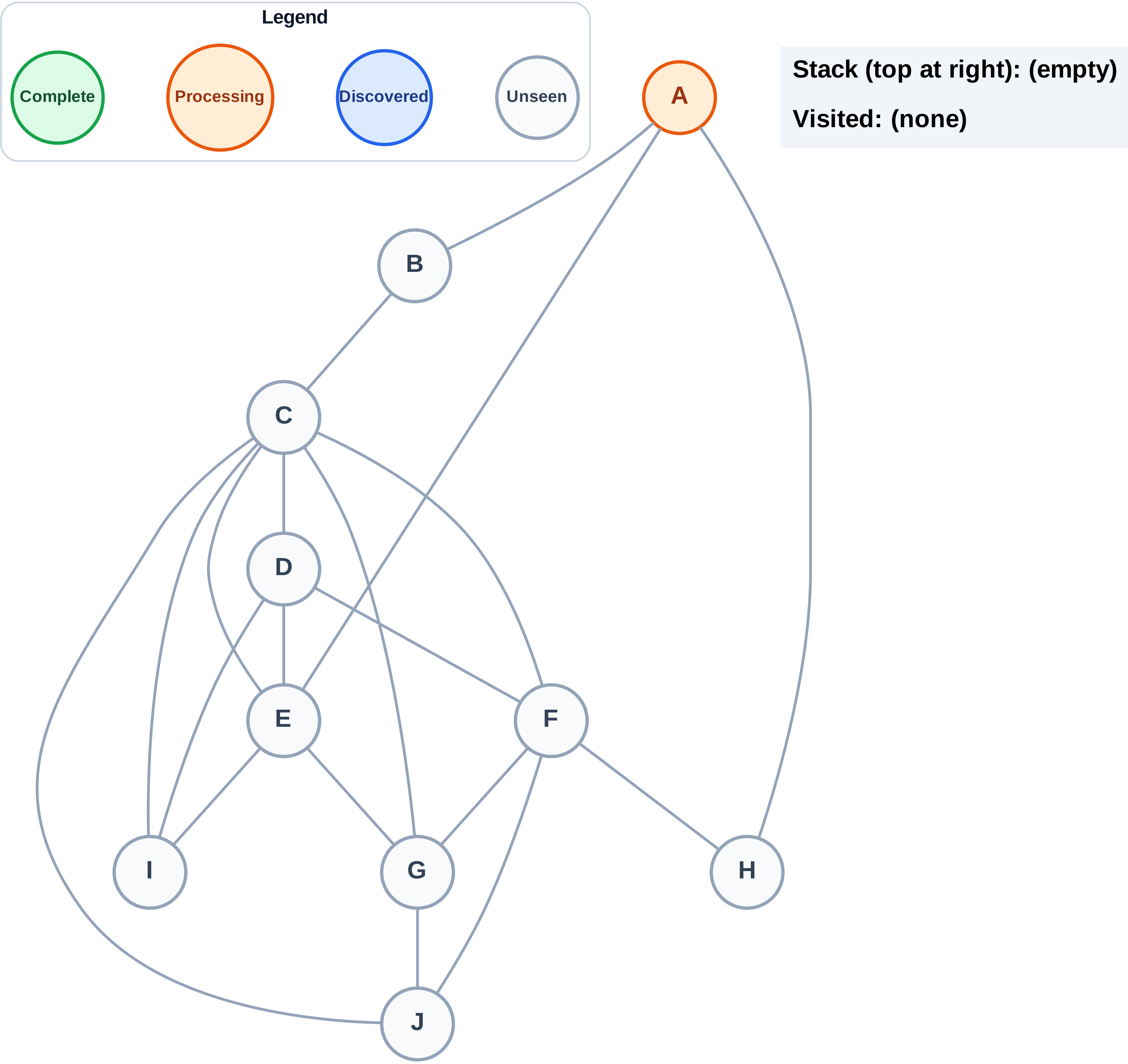
DFS Traversal

Step 1: Start at A



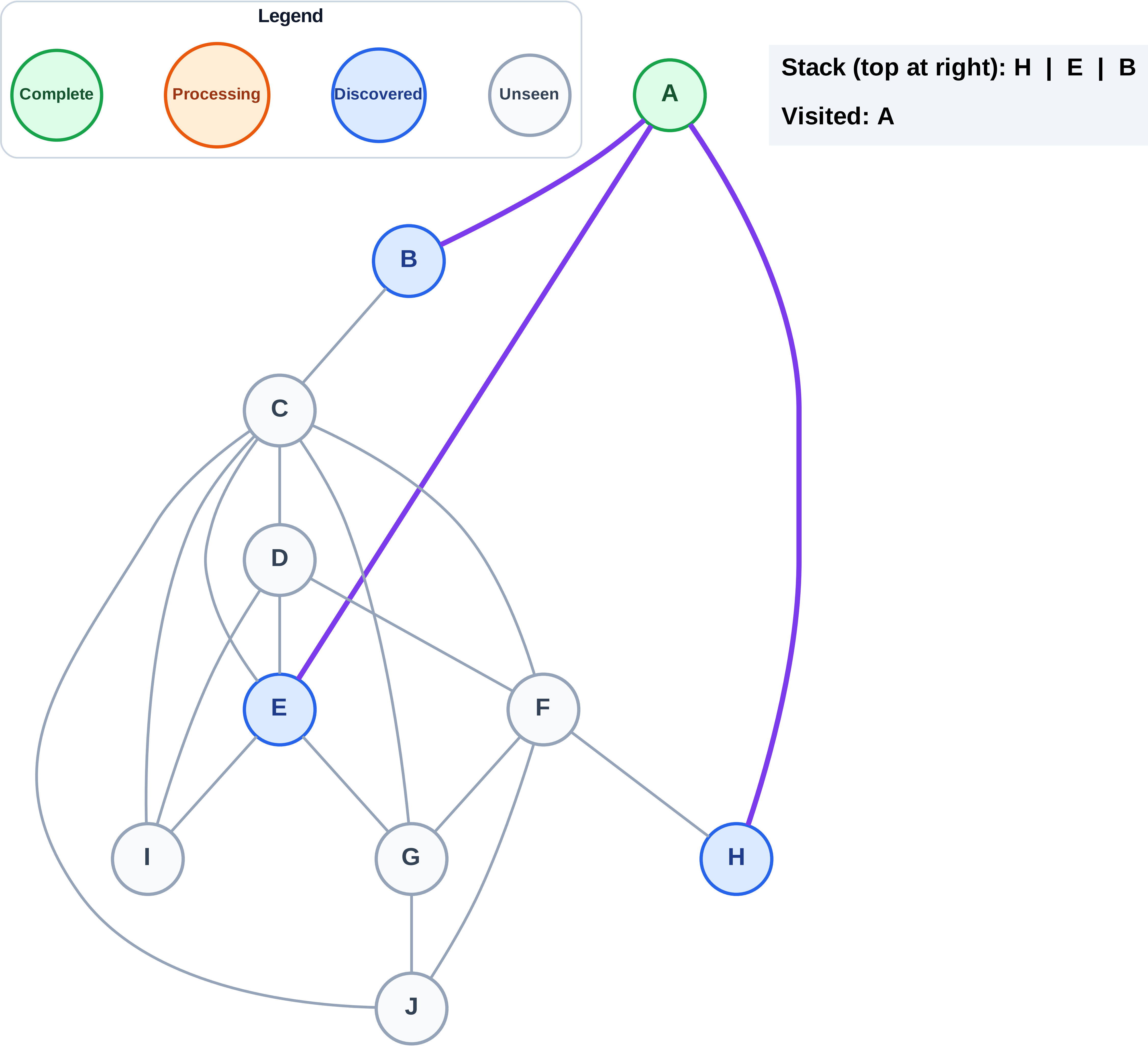
DFS Traversal

Step 2: Process node A



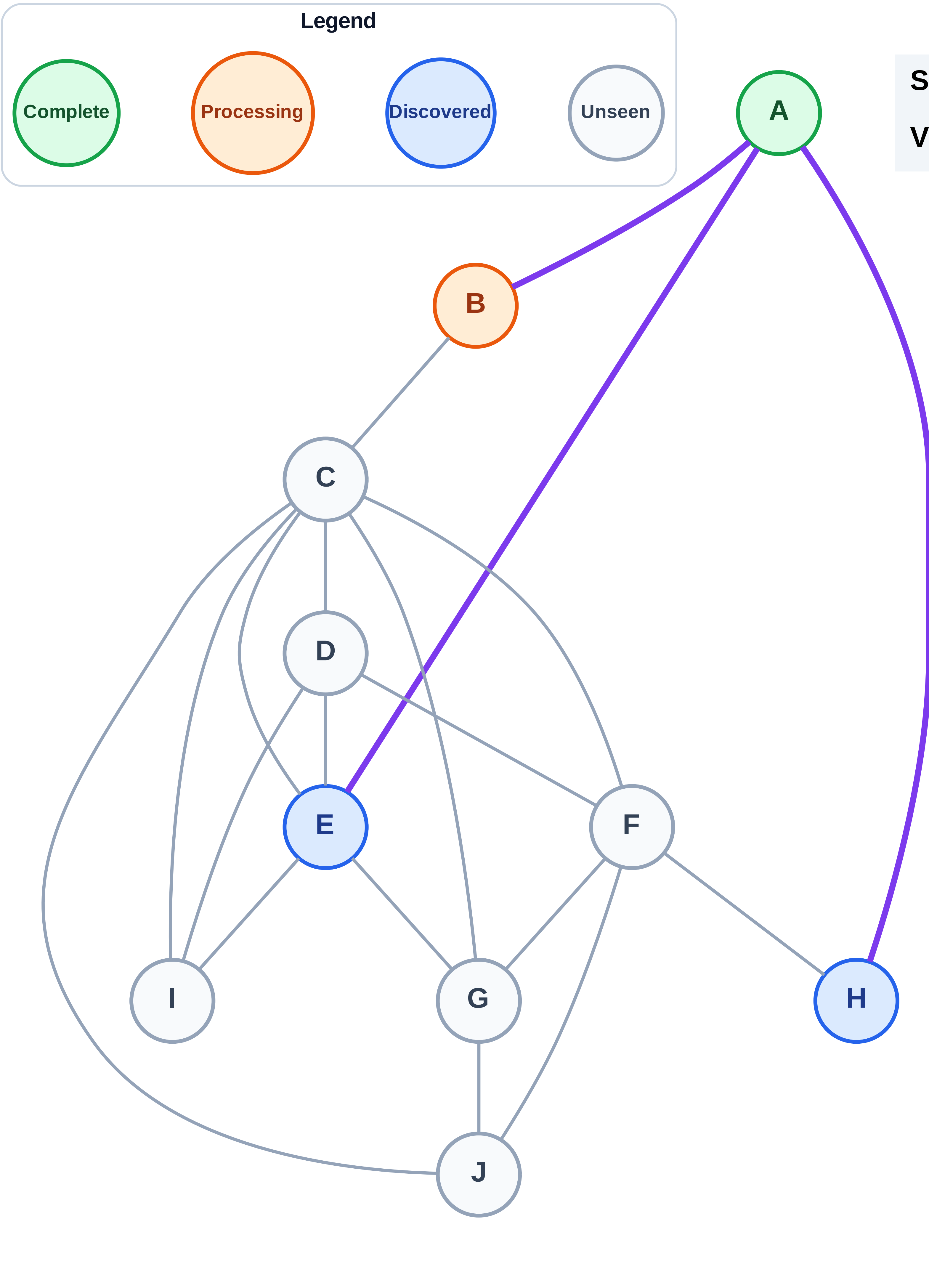
DFS Traversal

Step 3: Discover H, E, B from A



DFS Traversal

Step 4: Process node B

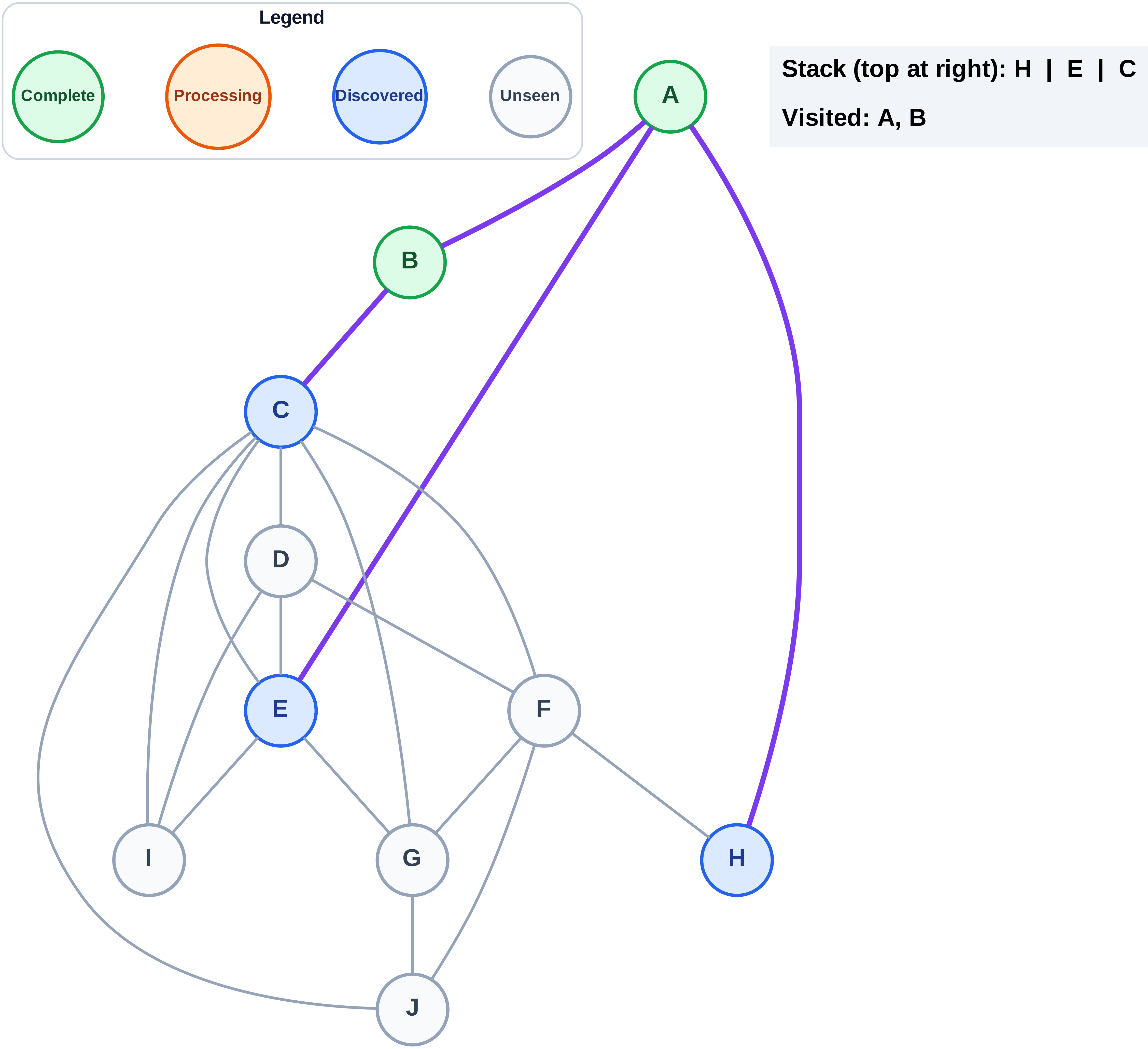


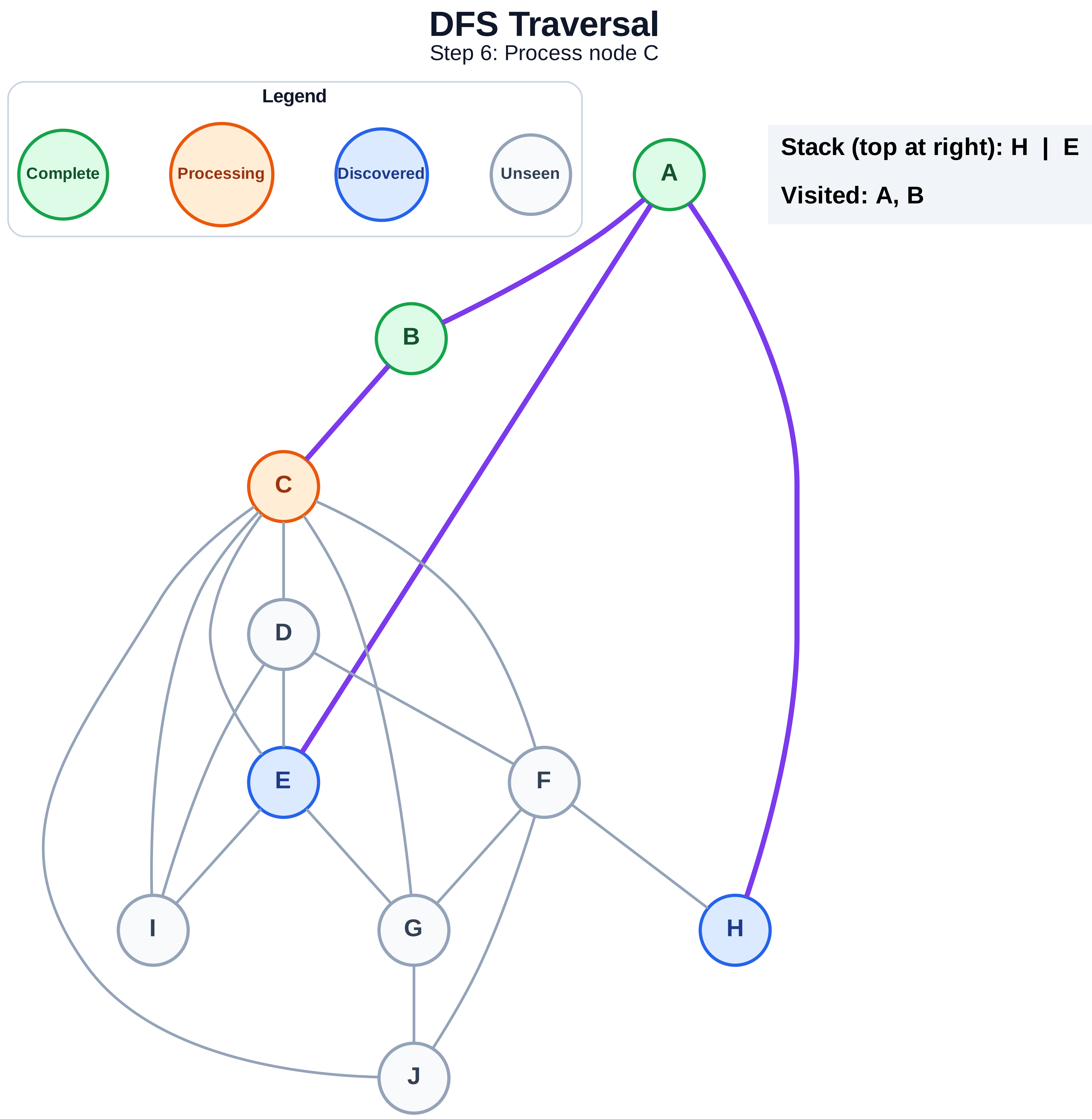
Stack (top at right): H | E

Visited: A

DFS Traversal

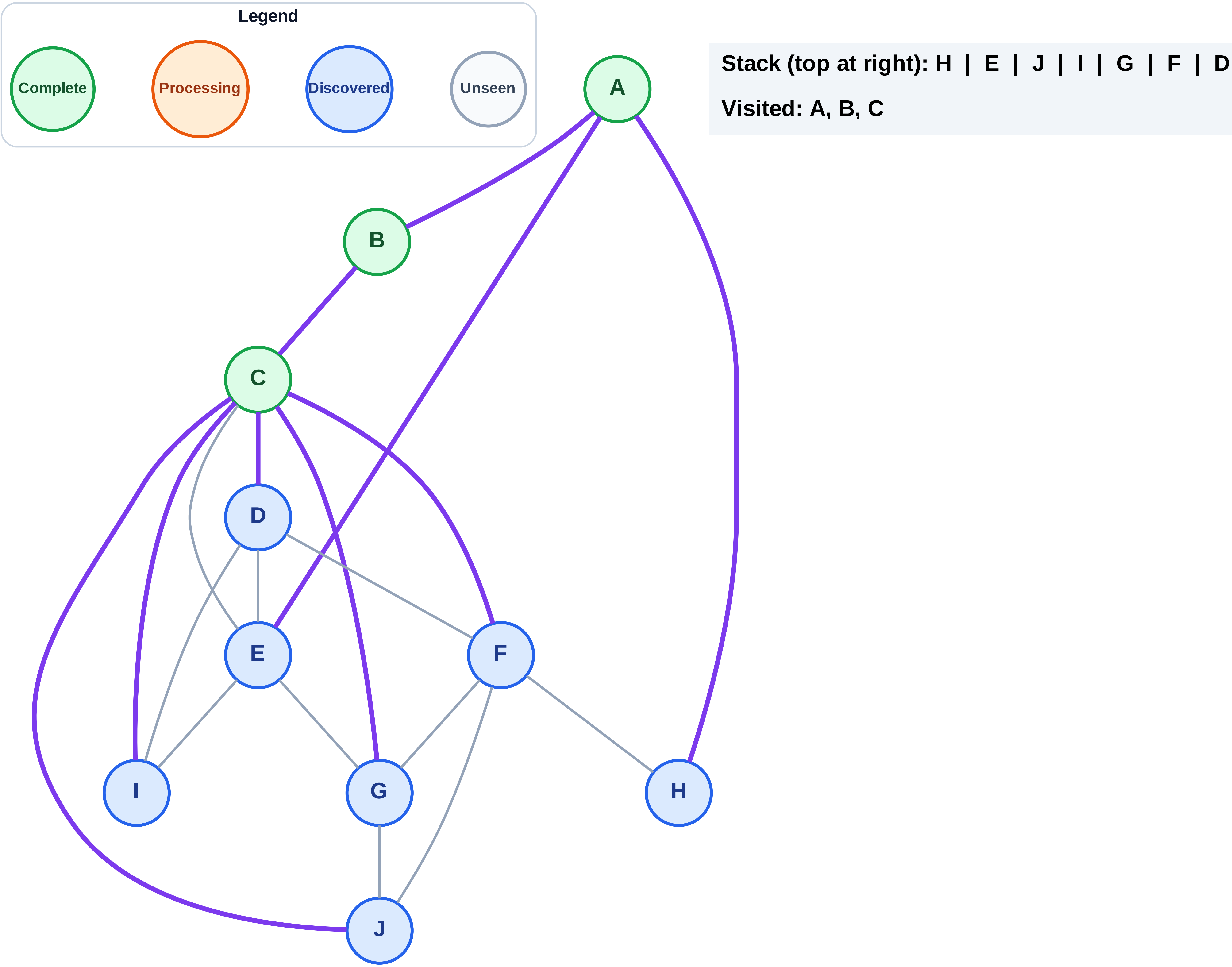
Step 5: Discover C from B




DFS Traversal
Step 6: Process node C
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | E
Visited: A, B
A
B
C
D
E
F
G
H
I
J

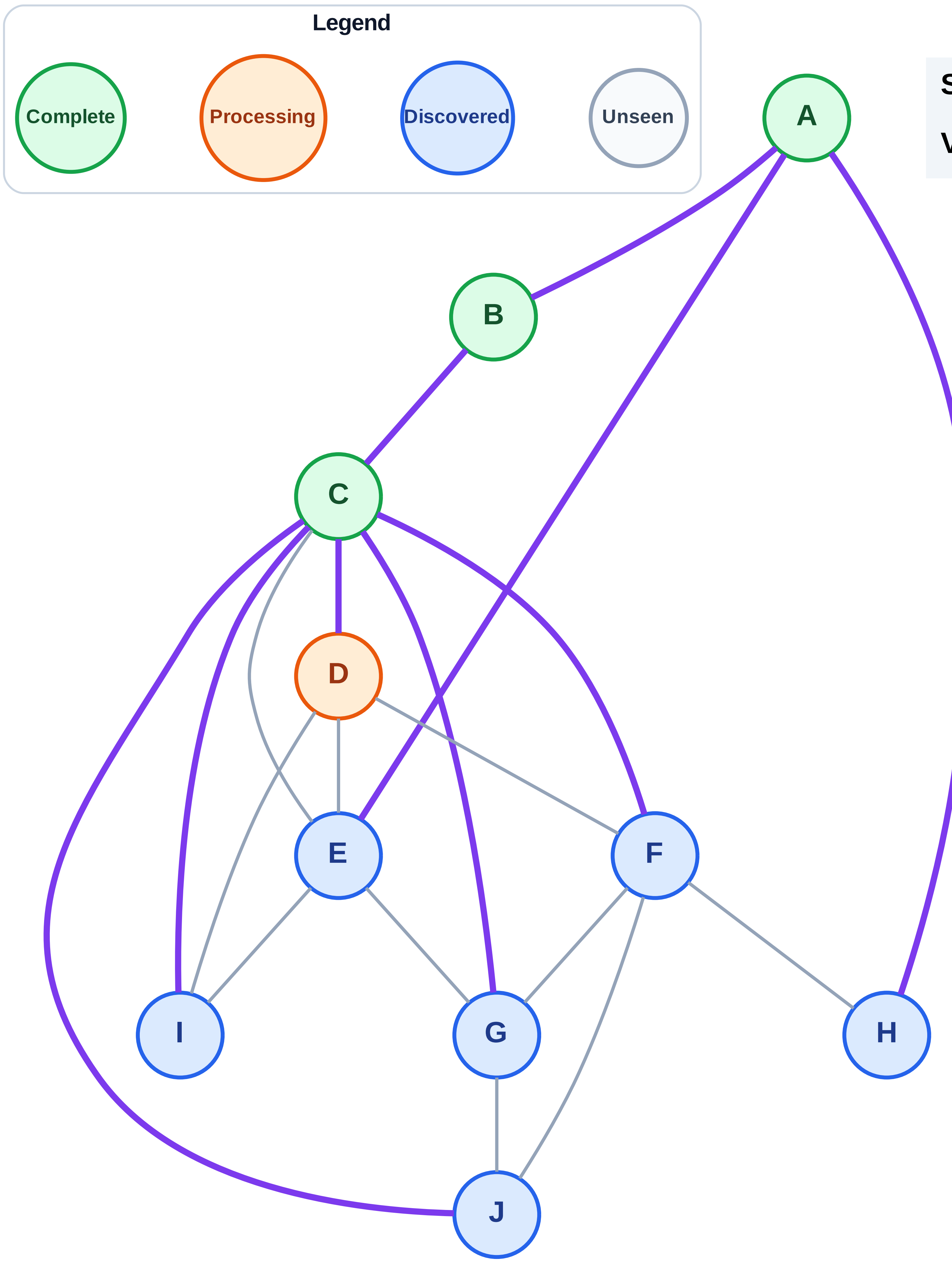
DFS Traversal

Step 7: Discover J, I, G, F, D from C



DFS Traversal

Step 8: Process node D

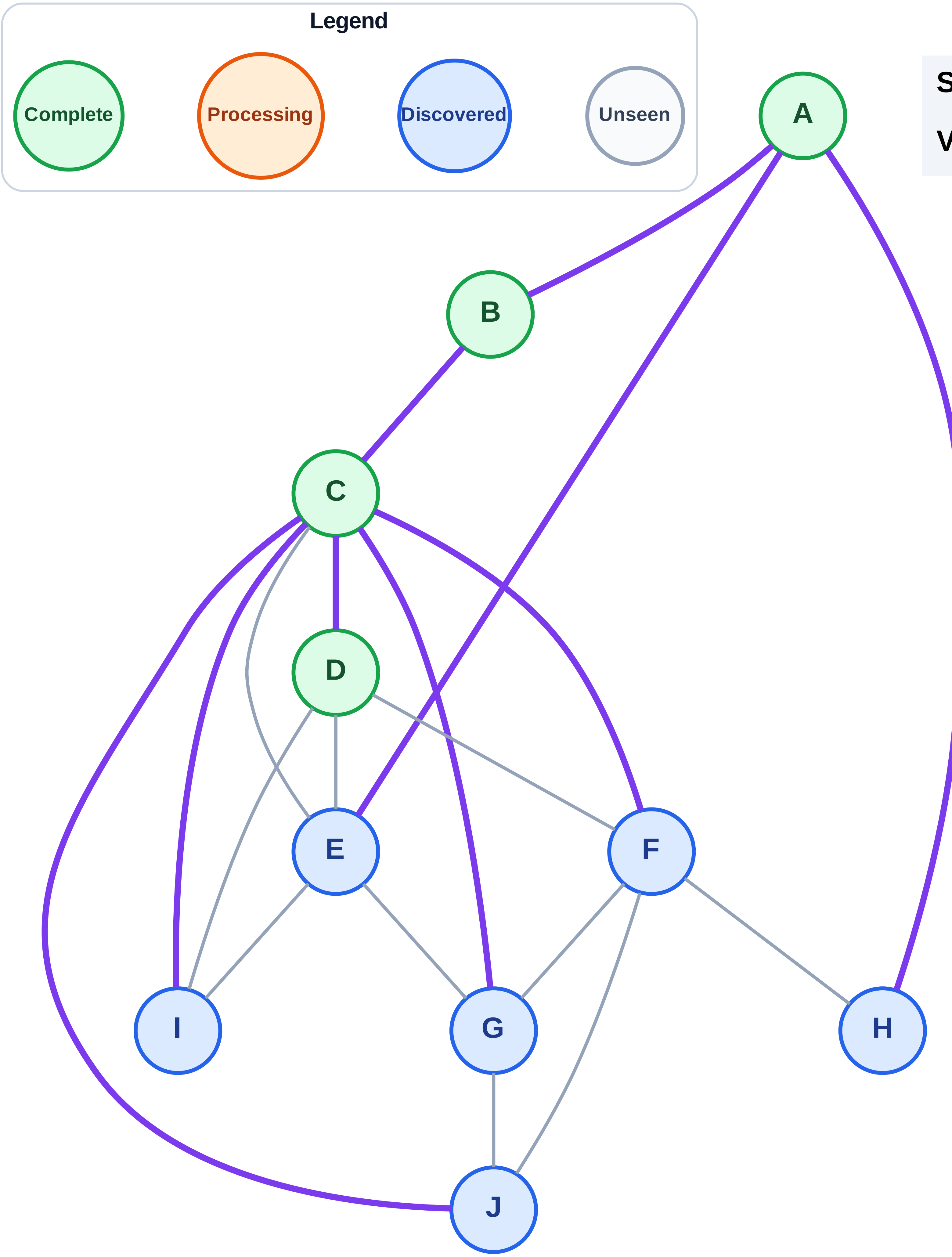


Stack (top at right): H | E | J | I | G | F

Visited: A, B, C

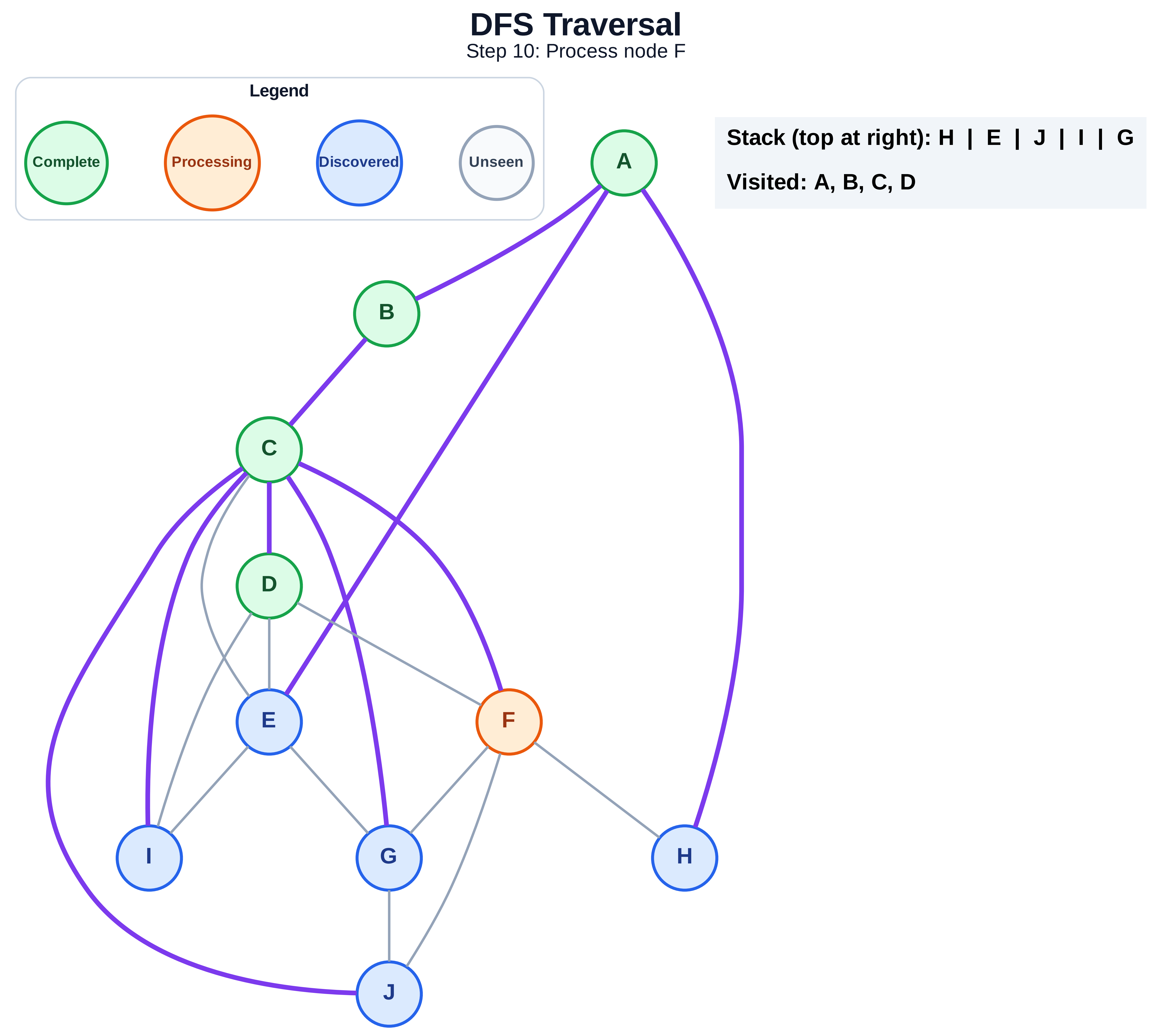
DFS Traversal

Step 9: No new neighbors from D



Stack (top at right): H | E | J | I | G | F

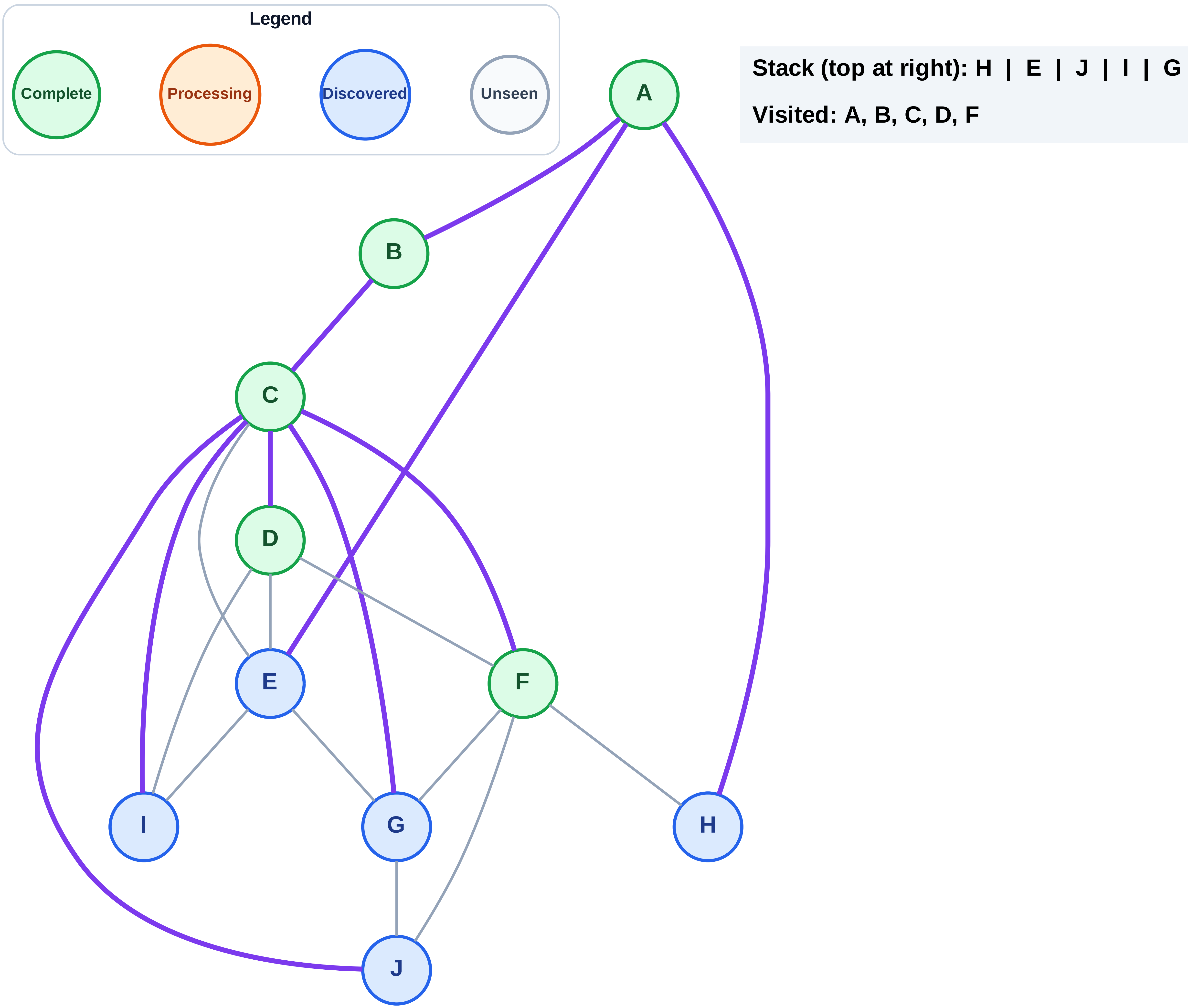
Visited: A, B, C, D



DFS Traversal
Step 10: Process node F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | E | J | I | G
Visited: A, B, C, D
A
B
C
D
E
F
I
G
H
J

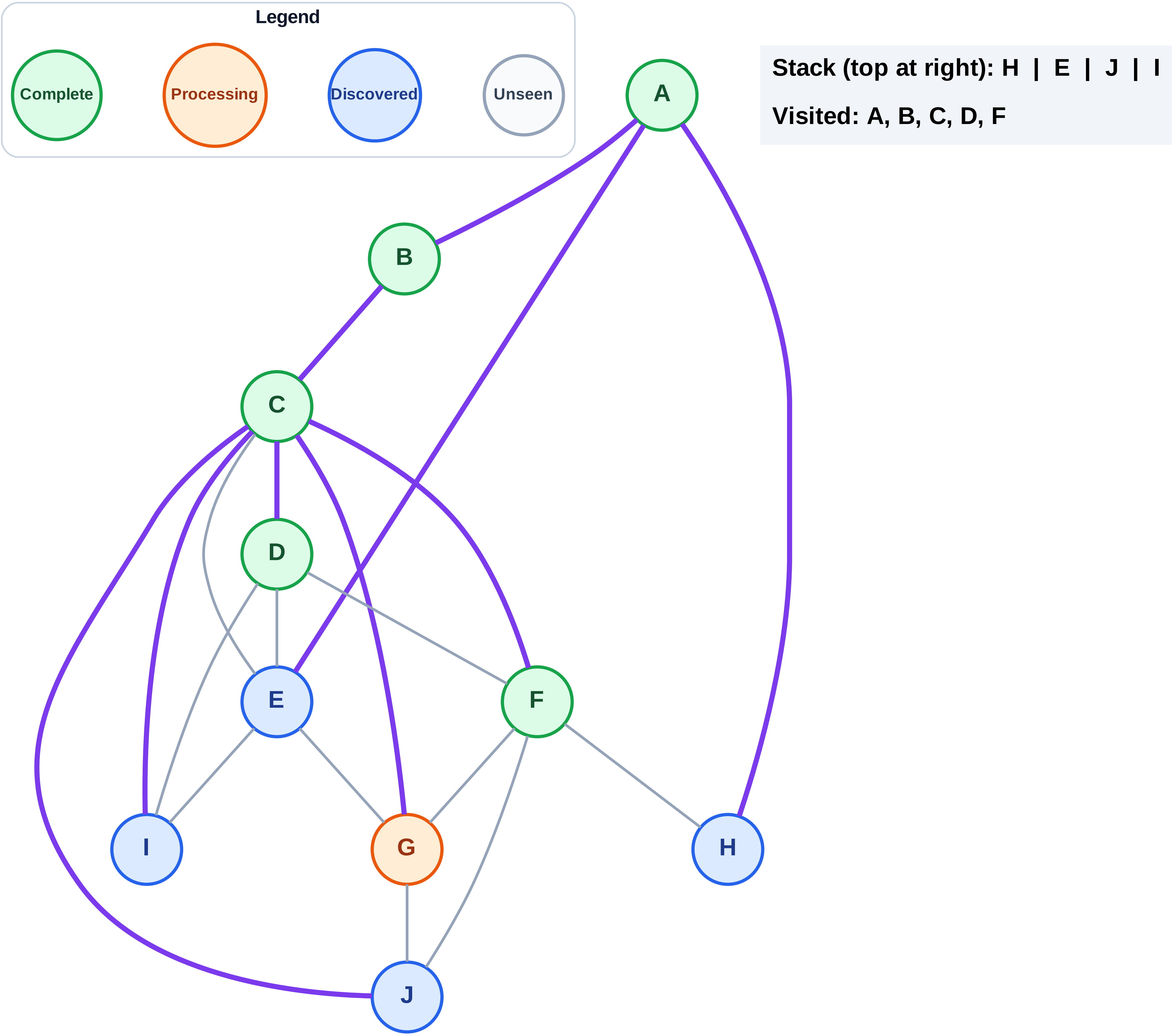
DFS Traversal

Step 11: No new neighbors from F



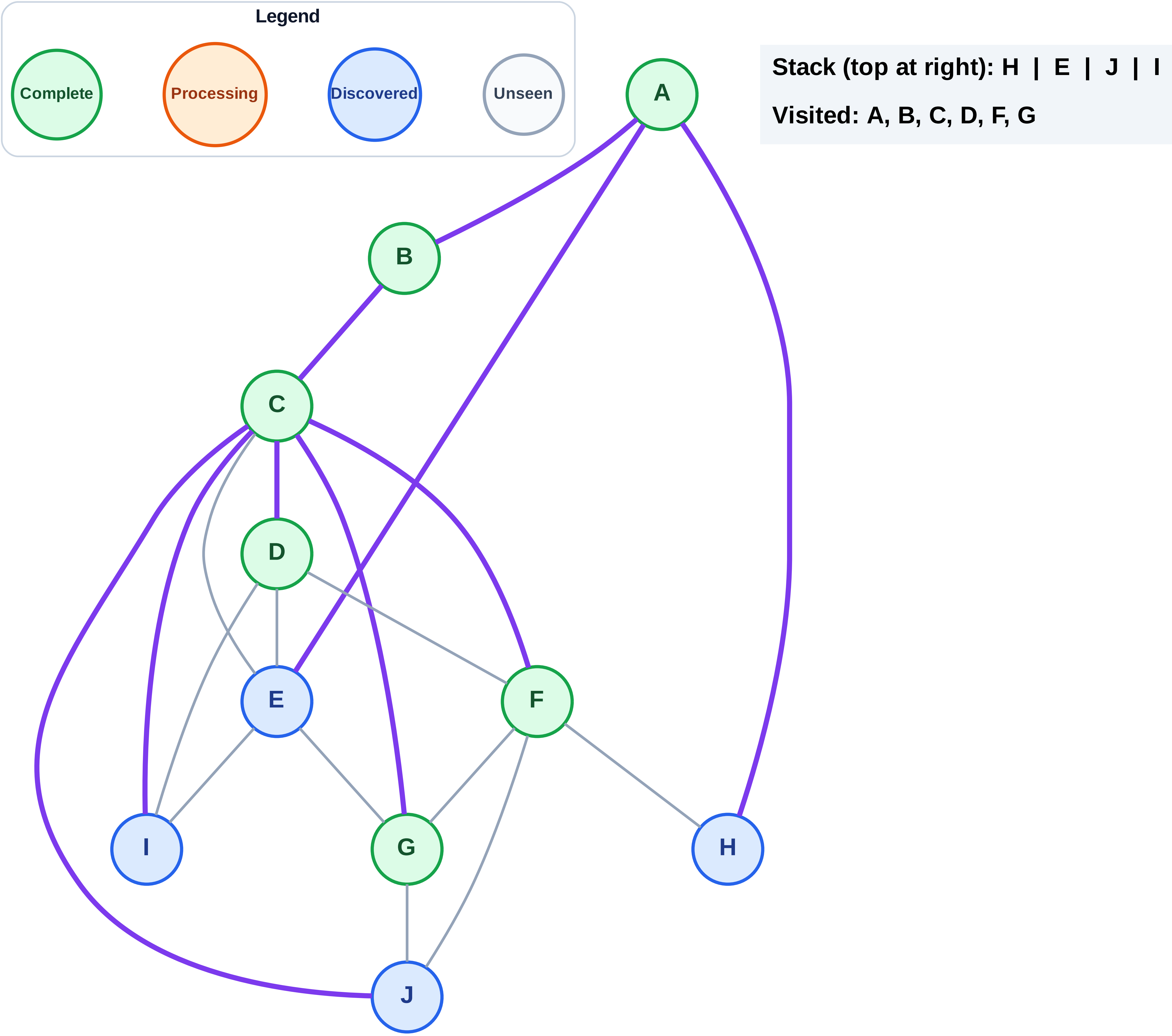
DFS Traversal

Step 12: Process node G



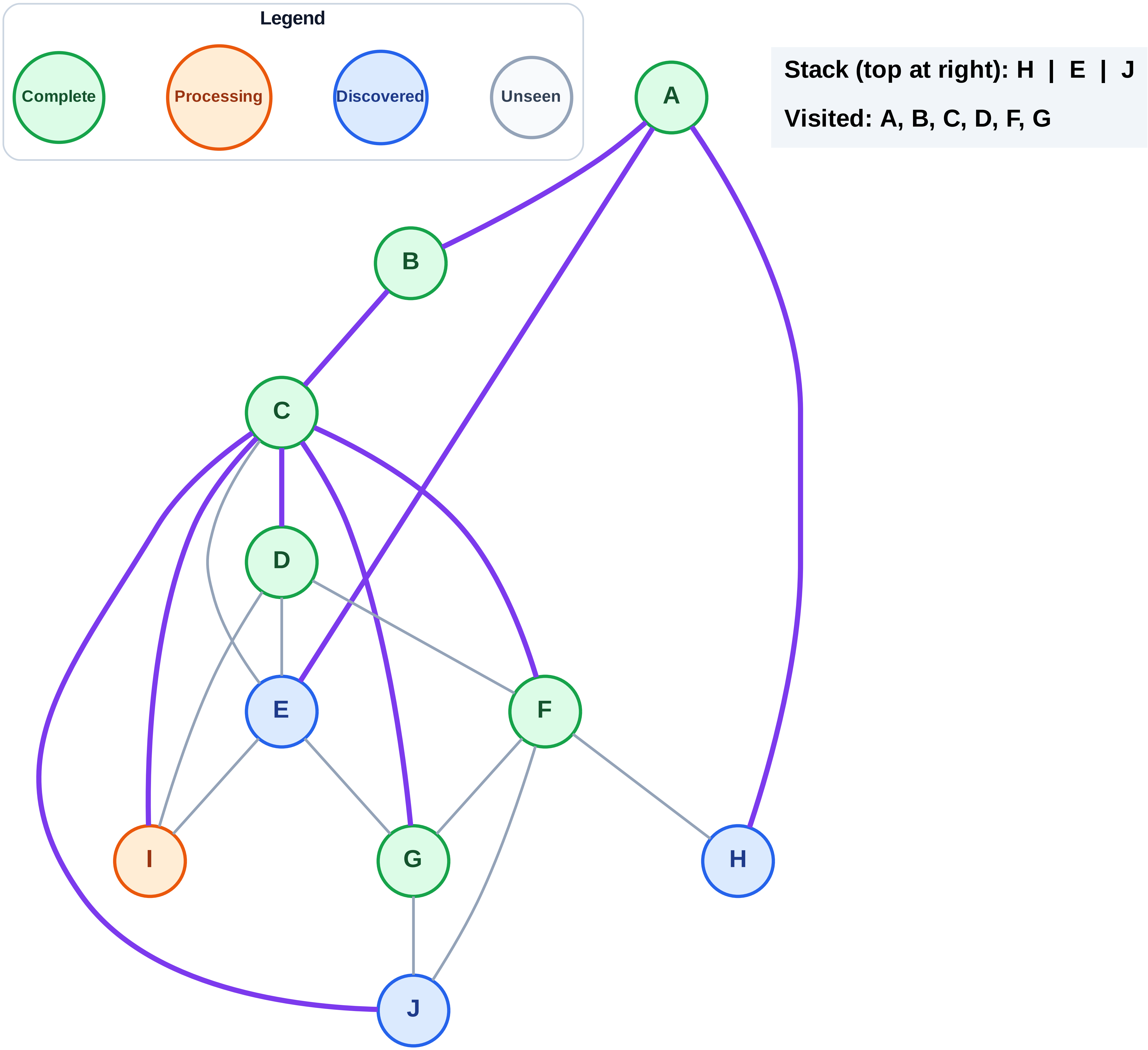
DFS Traversal

Step 13: No new neighbors from G



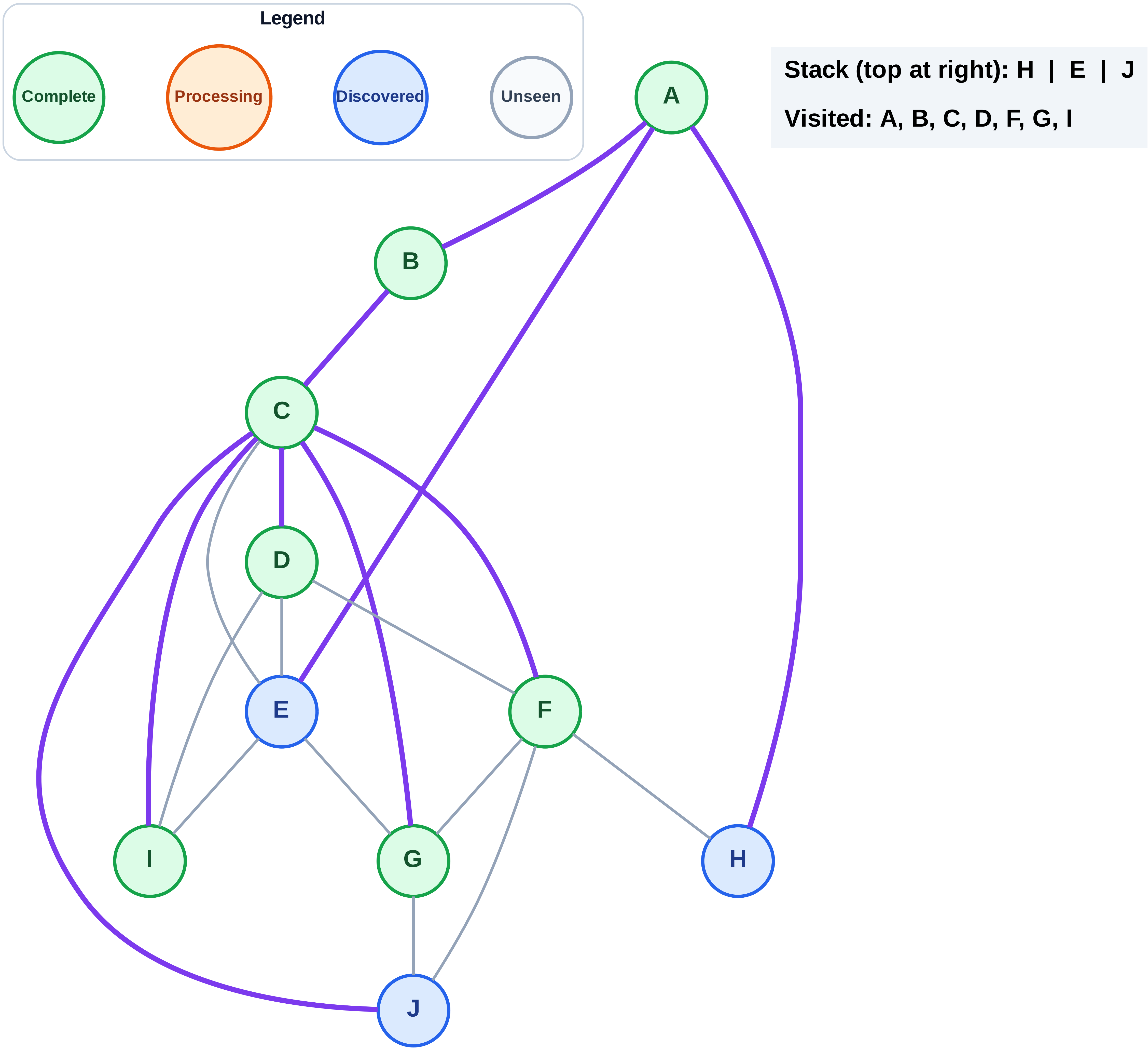
DFS Traversal

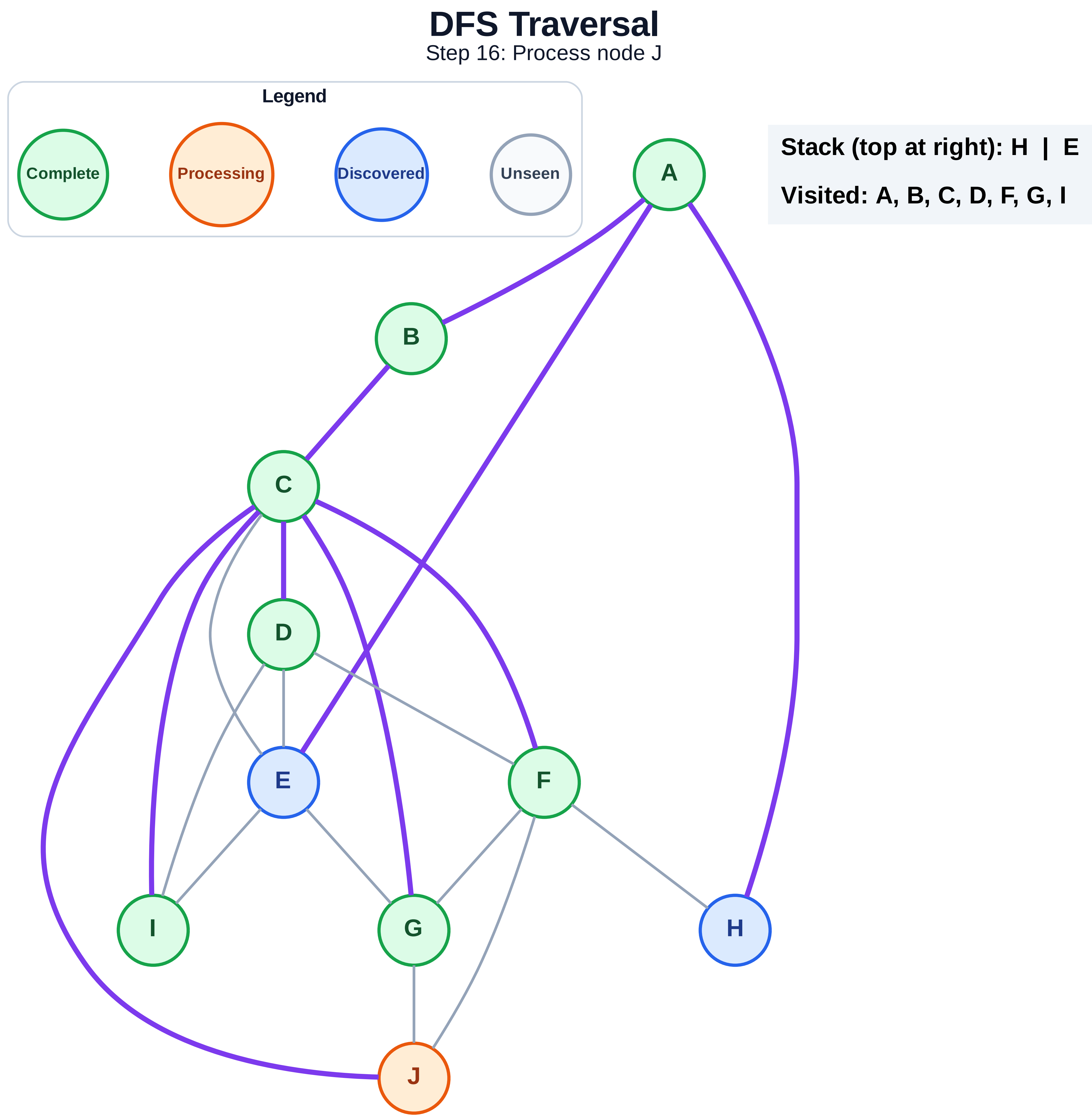
Step 14: Process node I

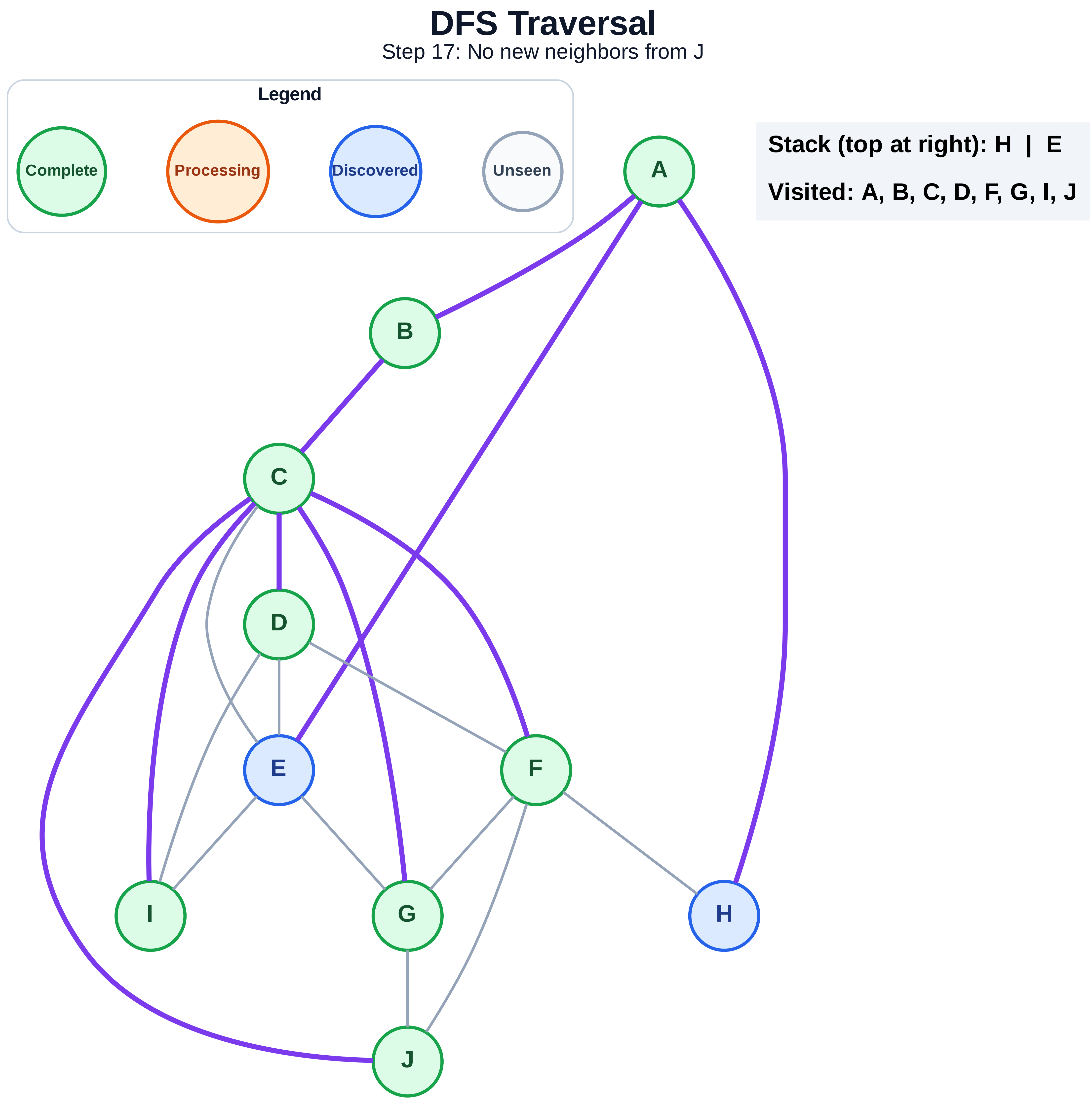


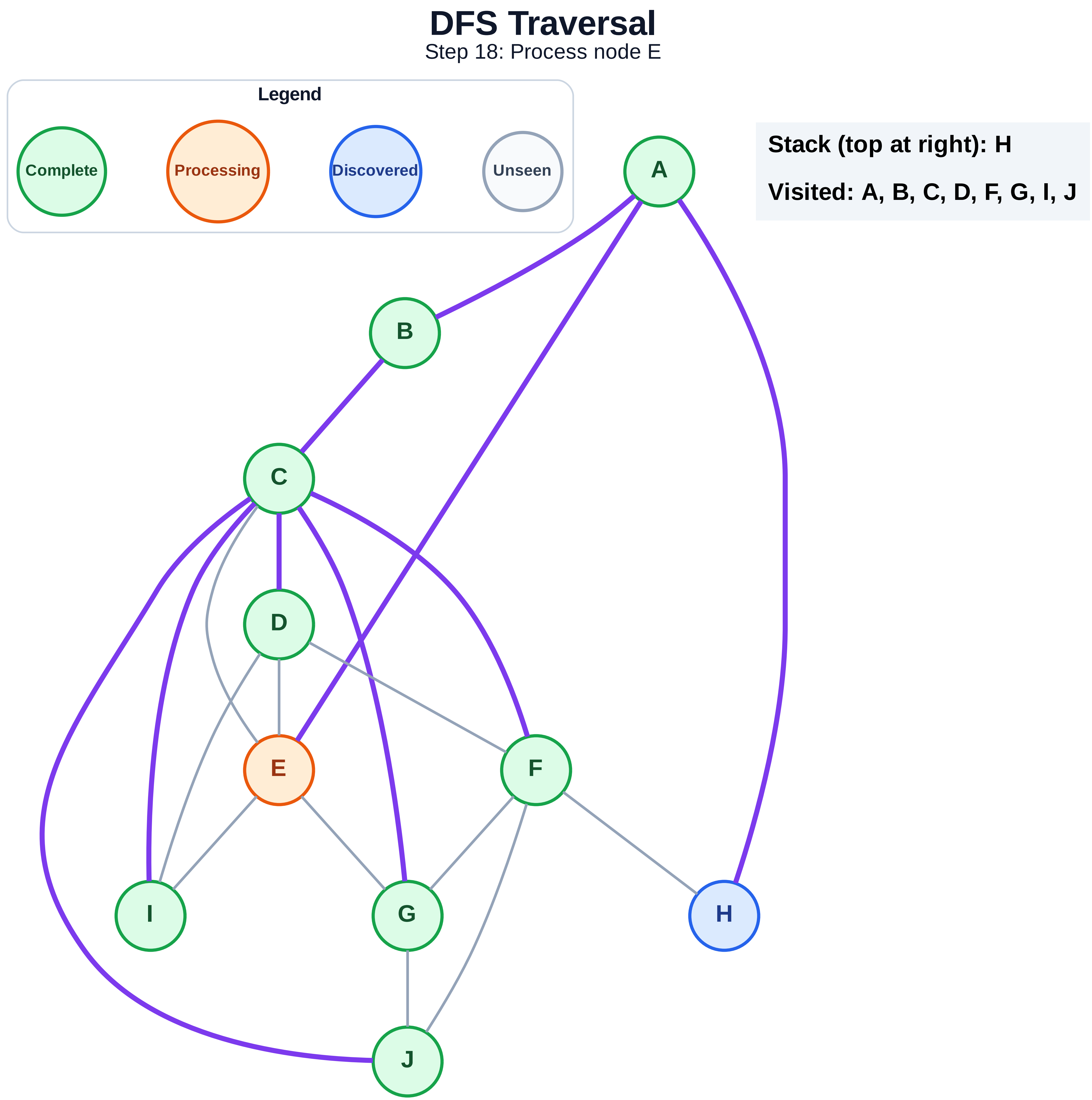
DFS Traversal

Step 15: No new neighbors from I



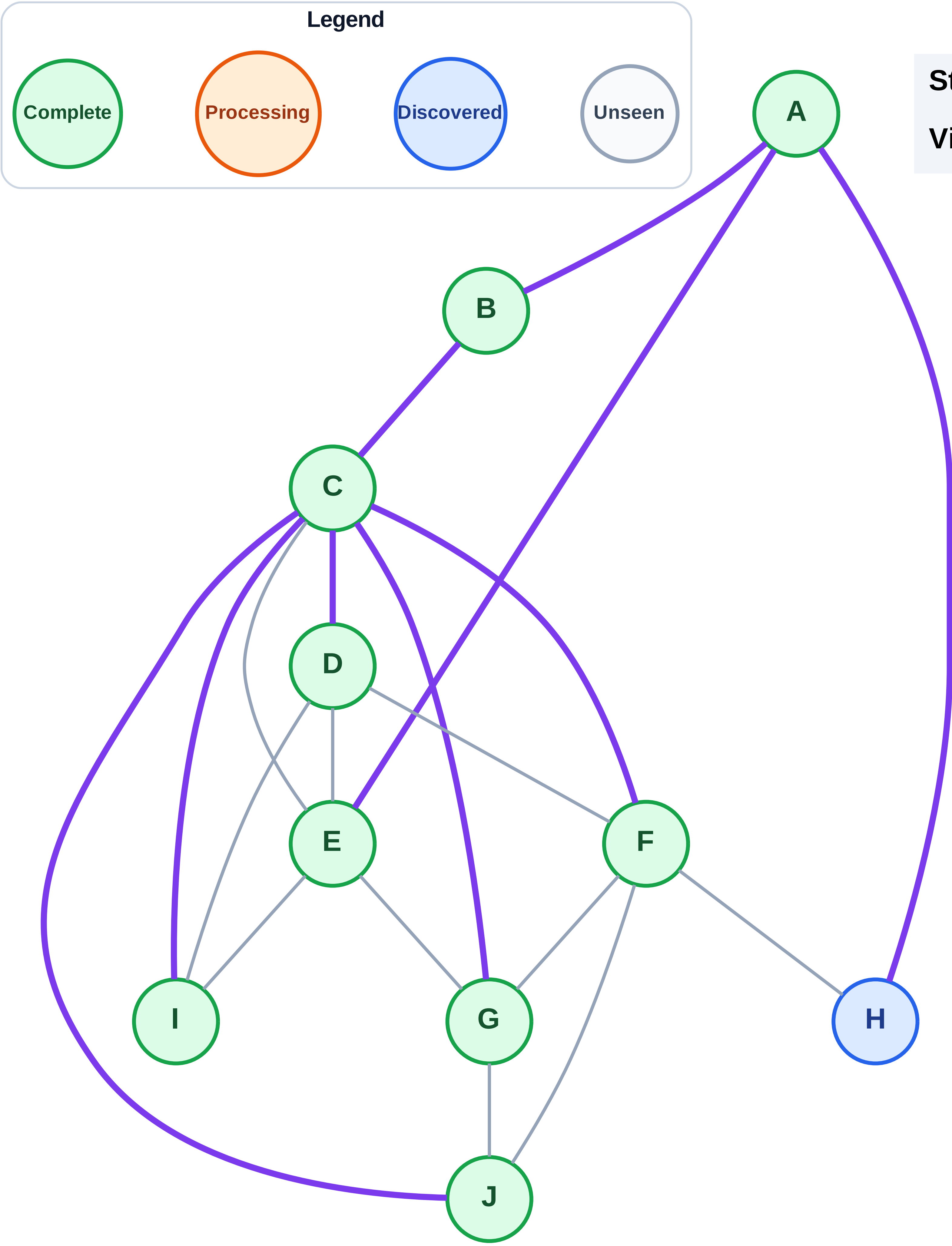

DFS Traversal
Step 16: Process node J
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | E
Visited: A, B, C, D, F, G, I
A
B
C
D
E
F
G
H
I
J


DFS Traversal
Step 17: No new neighbors from J
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | E
Visited: A, B, C, D, F, G, I, J
A
B
C
D
E
F
G
H
I
J


DFS Traversal
Step 18: Process node E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H
Visited: A, B, C, D, F, G, I, J
A
B
C
D
E
F
G
H
I
J

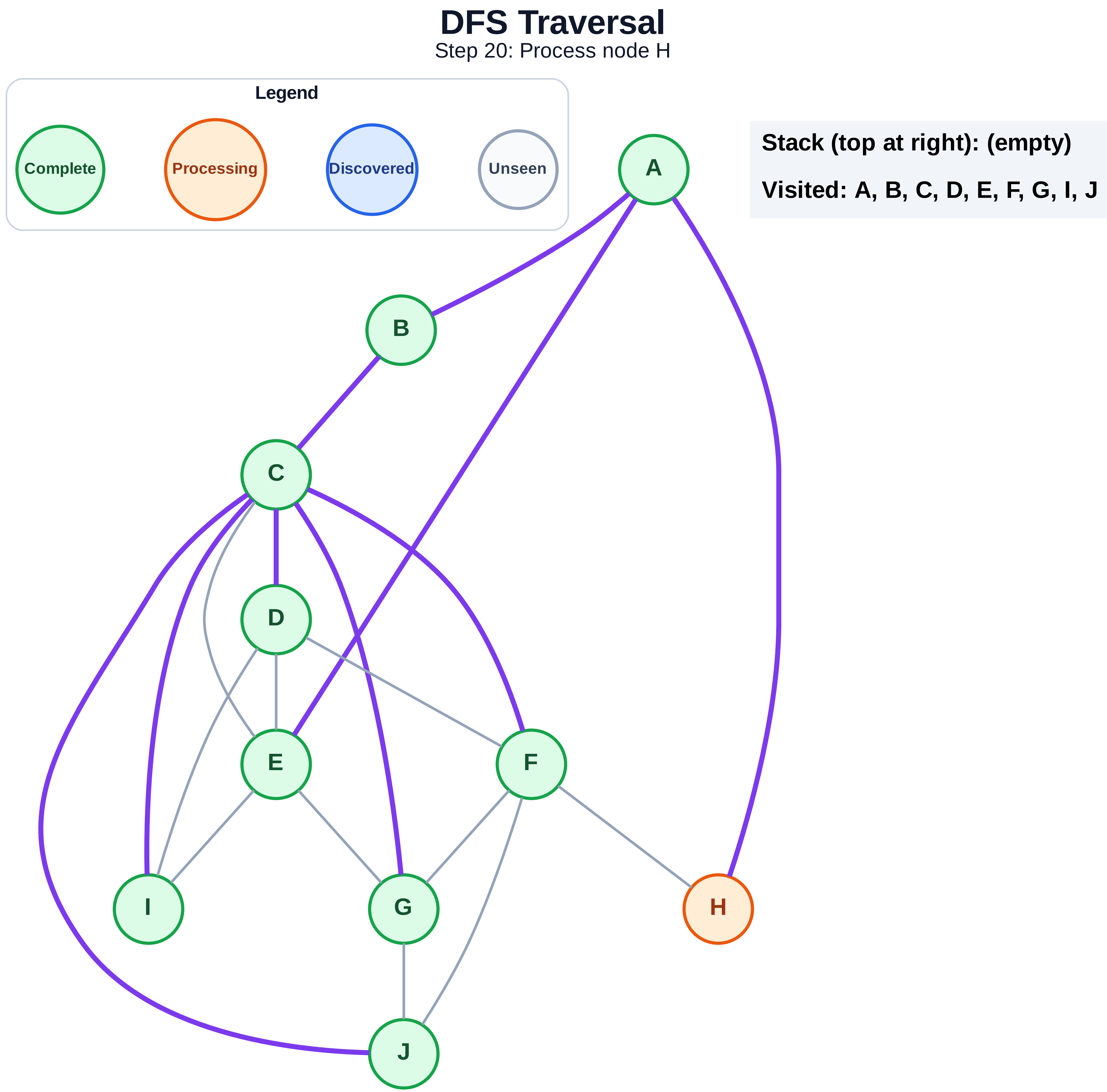
DFS Traversal

Step 19: No new neighbors from E



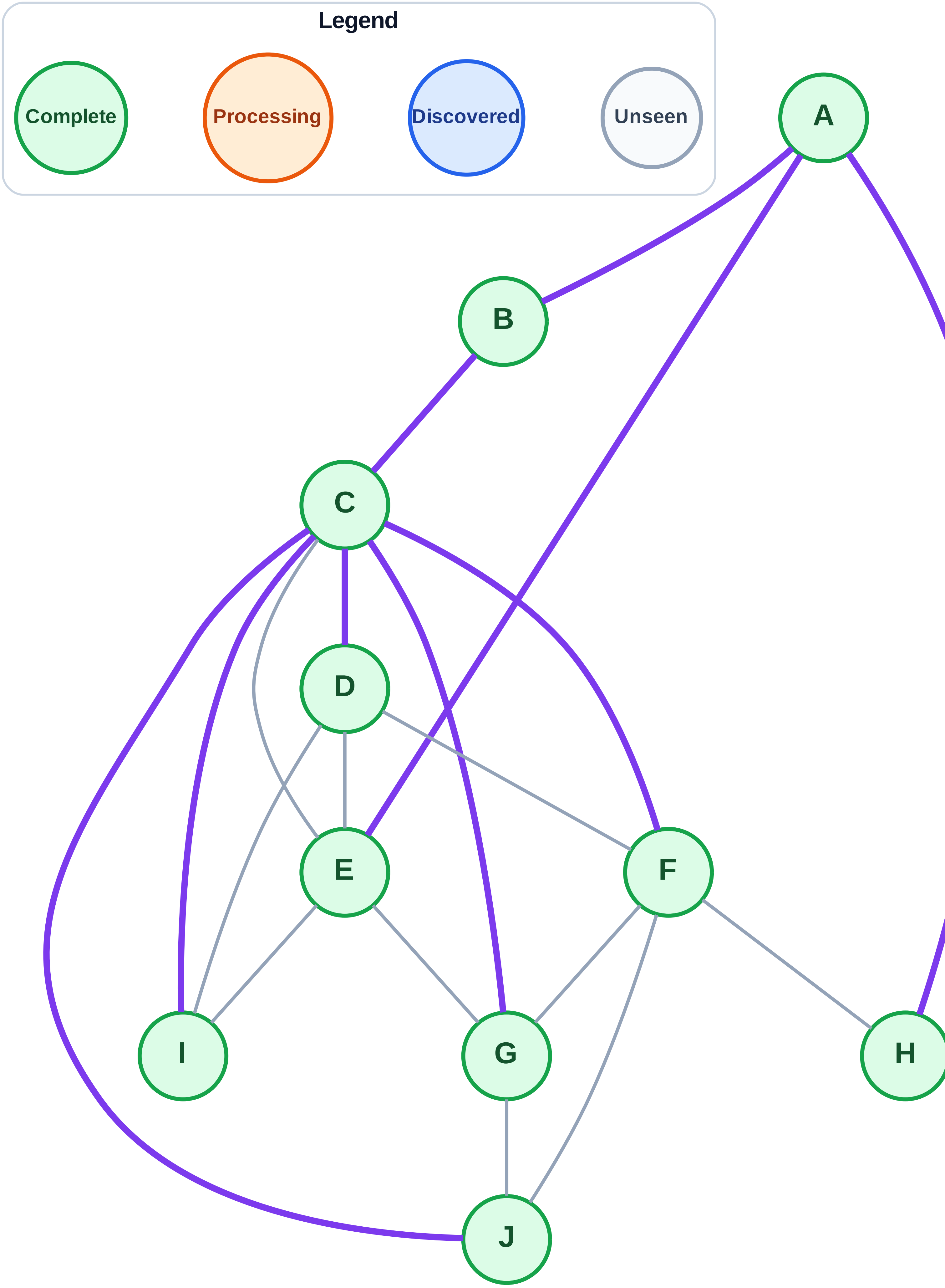
Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J


DFS Traversal
Step 20: Process node H
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, I, J
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 21: No new neighbors from H



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J